THE ACCOUNTING REVIEW
Vol. 91, No. 2
March 2016
pp. 677–705

American Accounting Association
DOI: 10.2308/accr-51182

The Impact of Audit Completeness and Quality on Earnings Announcement GAAP Disclosures

Joseph H. Schroeder
Indiana University

ABSTRACT: This study examines the role of the external audit in management's decision about the amount of GAAP financial statement information to disclose in the annual earnings announcement. The earnings announcement is a key disclosure provided by public companies. Yet, there is no requirement that earnings announcements contain audited GAAP numbers; in fact, recent trends indicate that a majority of companies release earnings before the completion of year-end audit fieldwork. I predict and find that companies that wait until the audit is more complete at the earnings announcement date and receive higher quality audits provide more detailed balance sheet, cash flow statement, and overall GAAP disclosures. I also provide evidence that complete audits and higher quality audits impact the information content of the earnings announcement. The combined results indicate that audit completeness and quality help facilitate more detailed earnings announcement disclosures and have implications for the equity market.

Keywords: audit completeness; audit quality; earnings announcement; PCAOB Standards; GAAP disclosures.

Data Availability: Data are publicly available from sources identified in the text.

I. INTRODUCTION

The annual earnings announcement (EA) is an important disclosure that is provided by public companies. Due to its timely release and salience, investors respond more strongly to the EA than to the subsequent release of the 10-K filing (Beyer, Cohen, Lys, and Walther 2010). Furthermore, recent research suggests that EAs are more informative to investors when the announcement contains more detailed GAAP (generally accepted accounting principles) financial statement information (Francis, Schipper, and Vincent 2002; Collins, Li, and Xie 2009; Pawlewicz 2011; Miao, Teoh, and Zhu 2013).[1] Despite the attention EAs receive from investors, there is no regulation requiring an audit of the GAAP financial statement information presented within the disclosure.

I appreciate the support and guidance of my dissertation committee: Chris Hogan (Chair), Marilyn Johnson, Kathy Petroni, and Edward Fee. I appreciate helpful comments and suggestions from Clive S. Lennox (editor), two anonymous reviewers, Andrew Acito, Randolph Beatty, Scott Bronson, Matt DeAngelis, Mark Evans, Jere Francis, Matt Hoag, Karla Johnstone, Harold Kazanabon, Dan Lynch, Dara Marshall, Nathan Marshall, Adi Masli, Elaine Mauldin, Brian Mayhew, Miguel Minutti-Meza, Brian Miller, Mark Myring, K. Ramesh, Dahlia Robinson, Marcy Shepardson, Tyler Thomas, Dan Wangerin, Teri Yohn, and the workshop participants at George Mason University, Indiana University, Michigan State University, The Pennsylvania State University, University of Miami, University of Missouri, University of Southern California, University of Wisconsin–Madison, the 2012 Miami Rookie Conference, and the 2013 AAA Annual Meeting in Anaheim. This paper was made possible by the generous financial support of the Eston Research Fund while I was a Ph.D. student at Michigan State University.

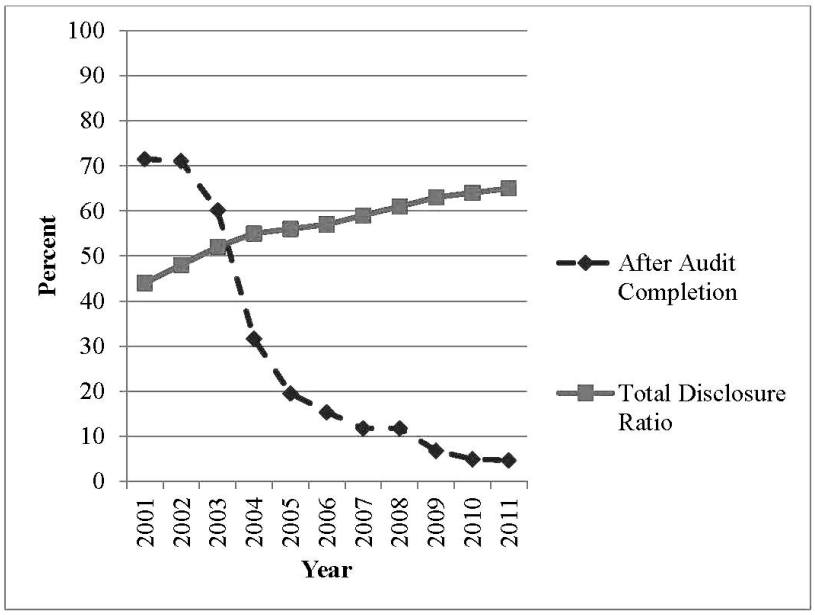
Editor's note: Accepted by Clive S. Lennox.

Submitted: August 2014
Accepted: June 2015
Published Online: June 2015

[1] Earnings announcement disclosures range from less detailed income statement summary disclosures to managers providing the full set of GAAP financial statements (i.e., income statement, balance sheet, and cash flow statement). See the Hampshire Group (CIK 887150) March 4, 2004 EA for an example of limited summary income statement disclosure (available at: http://www.sec.gov/Archives/edgar/data/887150/000088715004000003/0000887150-04-000003-index.htm). See the Delta Woodside Industries (CIK 806624) August 19, 2003 EA for an example of detailed GAAP disclosure (available at: http://www.sec.gov/Archives/edgar/data/806624/000101989203000111/0001019892-03-000111-index.htm)

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

FIGURE 1
EA in Relation to Audit Completion and Disclosure Ratio Trends



The above figure is based on the sample for years 2001 to 2011 (see Table 1). After audit completion is defined as earnings announcements that are released either on or after the audit report date. The total GAAP disclosure ratio is based on the D'Souza et al. (2010) earnings announcement GAAP disclosure ratios. The ratio is disclosed GAAP line items in the earnings announcement to line items that are ultimately disclosed in the 10-K filing. More discussion can be found in Section III and Appendix A.

Contrary to widespread belief that the financial statement information in the announcement is audited (Securities and Exchange Commission [SEC] 2002, 2003),[2] recent empirical research indicates that following the implementation of Public Company Accounting Oversight Board (PCAOB) Auditing Standards Nos. 2 and 3 (AS2/3) the majority of companies issue their annual EAs before the completion of the year-end audit (Krishnan and Yang 2009; Bronson, Hogan, Johnson, and Ramesh 2011).[3] At the same time, there has been an increasing trend of companies providing more GAAP financial statement details in the EA (Pawlewicz 2011; Miao et al. 2013). Figure 1 illustrates these trends, showing that the percentage of companies waiting until on or after the audit report date to release the EA has decreased from a high of 72 percent in 2001 to a low of 5 percent in 2011. At the same time, overall GAAP disclosures, measured using the disclosure ratio from D'Souza, Ramesh, and Shen (2010), increased 55 percent between 2001 and 2011.[4] This raises the question of the external auditor's role in management's decision to disclose GAAP financial statement information in the annual EA.

In this study, I examine whether audit completeness and audit quality are associated with the amount of GAAP financial statement information that management includes in their EA. This allows for a better understanding of the external audit's role in facilitating more detailed disclosures. Management may be more willing to disclose additional GAAP financial statement information, such as more details from the income statement, balance sheet, and/or cash flow statement, when there is a more complete audit at the EA release date and a higher quality audit, because management will have more confidence in the financial reporting system and the underlying financial statement balances. Alternatively, the auditor may have little effect on management's disclosure decisions because of existing regulations, the market demand for timely information, and the presence of alternative governance mechanisms that increase management's confidence in the financial statements.

[2] One of the key assumptions underlying the SEC Proposed Rule 33-8089 and Final Rule 33-8128, which reduced the 10-K and 10-Q filing deadlines for accelerated filers, is that the audit is "essentially complete" as of the EA release date. Specifically stated in the final rule, "We understand as a general matter that audit work is essentially completed and other steps have been taken to ensure the accuracy of the earnings announcement" (SEC 2002, 2003).

[3] Effective November 15, 2004, PCAOB Auditing Standard No. 2 established requirements for Section 404(b) internal control audits for accelerated filers, and Auditing Standard No. 3 established work paper documentation requirements for all registrants (PCAOB 2004a, 2004b).

[4] The D'Souza et al. (2010) disclosure ratio is the number of line items for all three financial statements reported in the EA per the Compustat Preliminary History File divided by the total line items that subsequently appear in the initial 10-K filing per the Compustat Unrestated Quarterly File.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

Prior research provides evidence that the information content of the EA is increasing in the amount of GAAP disclosure. Greater detail allows investors to better understand the implications of current performance for future performance (Francis et al. 2002; Collins et al. 2009; Pawlewicz 2011). Providing only net income in the EA makes it challenging for investors to determine the quality and valuation implications of earnings. By providing balance sheet and cash flow statement information in the EA, investors are able to parse out noise in accruals, leading to better forecasts of future earnings and cash flows (Chen, DeFond, and Park 2002; Baber, Chen, and Kang 2006; Wasley and Wu 2006; D'Souza et al. 2010; Miao et al. 2013).

Given that investors find more detailed EA GAAP disclosures to be informative, it is important to determine the role of the audit in management's decision to provide more detailed disclosures. Practitioner organizations recommend the inclusion of the external auditor in the EA disclosure review process as a "best practice." They also encourage member companies to wait until the audit is complete (or substantially complete) prior to releasing the EA (Bochner and Blake 2008; Diamond and Yevmenenko 2011; Deloitte 2012). Releasing the EA closer to or after the completion of year-end audit fieldwork reduces management's uncertainty about specific account balances and reduces the likelihood that the preliminary financial statement information included in the EA will change before the filing of the 10-K (Bronson et al. 2011).

Higher quality audits improve the quality of the financial reporting system through internal control testing and earlier interim procedures, as well as by resolving key financial statement issues more timely and accurately. A higher quality financial reporting system ensures that the financial information provided by that system is materially consistent with GAAP. Taken together, more complete audits and higher quality audits improve the quality of the financial reporting system and increase the accuracy of the financial statement outputs and the underlying account balances, thereby increasing management's confidence that the financial statement balances will not change between the earnings release and the filing of the 10-K. As a result, management is less exposed to litigation risk (Rogers and Van Buskirk 2009; Cutler, Davis, and Peterson 2013)[5] and reputation loss (Trueman 1986; Desai, Hogan, and Wilkins 2006; Yang 2012), which makes it less costly to release more detailed financial statement information.

On the other hand, existing regulations and market demand for more timely information suggest that the audit may not play a role in management's EA disclosure decisions. There is no SEC regulation requiring an audit of the information included in the EA; thus, management may release the preliminary financial statement information marking it as "unaudited" (Bochner and Blake 2008; Steinberg 2009; Marshall, Schroeder, and Yohn 2015). Furthermore, managers who feel pressure to meet the market's demand for timely information may not be willing to wait for the completion of audit fieldwork (Krishnan and Yang 2009; Bronson et al. 2011). Instead, these managers may increase their confidence in the historical GAAP information provided in the EA by making investments in alternative governance mechanisms, such as effective internal controls, a competent and objective internal audit function, and/or an expert audit committee, substituting for the role of the external audit.

Despite these possibilities, I predict that audit completeness and audit quality will be positively associated with the amount of GAAP financial statement information disclosed in the annual EA. I measure audit completeness as the number of days between the EA date and audit report date, with higher (lower) values indicating more (less) complete audits at the EA date. I operationalize audit quality using measures of both auditor size/resources and excess audit fees. Higher quality audits are more likely to be performed by larger audit firms that possess more resources and expertise (DeAngelo 1981; DeFond and Zhang 2014). Companies that desire higher levels of assurance over their financial statement balances would pay higher-than-expected audit fees to ensure that the audit firm can complete its job without making trade-offs between engagement profitability and audit quality (Blankley, Hurtt, and MacGregor 2012; Ball, Jayaraman, and Shivakumar 2012). I measure the level of detail provided in the EA by calculating the ratio of financial statement line items in the EA to the number of line items included in the 10-K (D'Souza et al. 2010).

Using a sample of 22,278 annual EA disclosures from 2001 to 2011, I find that the level of detail provided in the EA is positively associated with more complete audits and higher quality audits, after controlling for other factors affecting the level of disclosure details.[6] These results are driven primarily by detailed balance sheet and cash flow statement disclosures. The results are also economically significant as increases in the inter-quartile range of the audit completeness and quality proxies increase (relative to the mean) total GAAP, balance sheet, and cash flow statement ratios by 14.41, 14.41, and 43.36 percent,

[5] The historical GAAP financial statements presented in the earnings announcement are subject to the antifraud provisions of Section 10(b) and Rule 10b-5 of the Securities Exchange Act of 1934 (Bochner and Blake 2008; Steinberg 2009). Specific to the quantity of disclosures provided by companies, Cutler et al. (2013) find that companies with greater disclosures during a litigation class action period are more likely to receive significant settlements against the company. Furthermore, revisions of GAAP financial results reported in the earnings announcement could result in management/companies being named in a class action lawsuit. A recent example is *Zhang v. Groupon* (2012), in which Groupon was accused of filing false and misleading financial results in the Q4 2012 earnings announcement. Specifically, Groupon announced on March 30, 2012, a downward revision of $0.04 per share to their previously issued February 8, 2012 earnings announcement due to revenue recognition concerns identified by their auditor (Stempel 2012; *Zhang v. Groupon* 2012).

[6] Consistent with D'Souza et al. (2010), I include measures for size, information demand/environment, litigation risk, bad news, filer characteristics, earnings announcement timing, industry effects, and year effects to control for other determinants of disclosure details in the earnings announcement.

American Accounting Association

respectively. These results are generally robust to propensity score matching and first-differencing research designs that control for observable and unobservable characteristics that may affect both the decision to release earnings prior to audit completion, audit firm choice, and the level of GAAP disclosures. However, I acknowledge that these additional robustness tests cannot completely rule out endogeneity concerns.

I also examine market implications of more detailed EA GAAP disclosures by examining cross-sectional differences in EA information content measured as abnormal stock returns, volume, and stock price volatility. I confirm prior research that documents that EAs are more informative when they have more detailed GAAP disclosures (e.g., Francis et al. 2002; Collins et al. 2009) and when they have complete or higher quality audits (e.g., Marshall et al. 2015; Teoh and Wong 1993). I extend this literature by providing evidence that more complete (higher quality) audits attenuate (enhance) the information content of more detailed EA GAAP disclosures. Further analysis suggests that the attenuating effect of complete audits on EA GAAP disclosures is due to these EA releases being less timely, with market participants likely inferring some of the GAAP information from earlier alternative information sources. The combined results further our understanding of how the interaction between audit completeness/quality and EA GAAP disclosure detail impacts equity market participants.

This study contributes to the accounting literature in several ways. Prior research on the role of auditing with voluntary disclosure is limited to finding that higher audit fees are associated with higher quality management forecasts (i.e., frequency, precision, timing, and accuracy); engaging an industry specialist audit firm is associated with higher Association for Investment Management and Research (AIMR) disclosure ratings; and EA revisions are less likely when the audit is complete at the release date (Dunn and Mayhew 2004; Bronson et al. 2011; Ball et al. 2012). I extend these studies by providing evidence that audit completeness and quality impact management's EA GAAP disclosures. The results support a complementary relationship between auditing and disclosure quality, answering the call of Ball et al. (2012) for more research examining the impact of auditing in other key disclosure settings.

There is an emerging stream of literature examining the implications of audit completeness as of the earnings release date. Marshall et al. (2015), examining the pre/post-AS2/3 period, provide evidence that information asymmetry is lower and market reliance is higher for earnings releases with completed audits. Bronson et al. (2011) find that complete audits lead to more reliable EA disclosures, and Bronson, Masli, and Schroeder (2015) find that more complete audits result in lower audit risk, which, in turn, reduces both audit fees and auditor turnover and increases audit quality. While these studies focus on the consequences to equity investors and the auditor/client dynamic, this study provides evidence that audit completeness has implications for management's GAAP disclosure decisions.

Finally, the results of this study should be of interest to regulators, especially the PCAOB. In a speech on December 5, 2011, PCAOB Chairman James R. Doty stated, "If the auditor's role expands into more relevant involvement of the attest function in the quarterly reporting and earnings release process, what is the potential for *report delay* and is that retardation effect justified by a benefit to investors" (PCAOB 2011; emphasis added). The findings herein suggest that the external audit plays an important role in key financial statement disclosures that are not subject to formal audit requirements. Given that the market places greater reliance on the EA (Beyer et al. 2010), regulators (especially the PCAOB) may need to consider the timeliness of audits with respect to annual earnings releases when assessing auditing standards.

The next section develops the hypotheses. Section III discusses the research methodology and sample selection. Sections IV and V present the main results and market implications analyses, respectively. Section VI concludes the paper.

II. LITERATURE REVIEW AND HYPOTHESES

Earnings Announcement Disclosures

The annual EA is a key information source for investors as it provides the initial benchmark for assessing the accuracy of earlier management forecasts and other information sources (Beyer et al. 2010). Furthermore, investors value the information in the EA more than that included in subsequently issued periodic filings (i.e., 10-K), as evidenced by higher market reactions, due to the EA's timelier release and the salience of the disclosure (Stice 1991; Kothari 2001; Li and Ramesh 2009). Although the EA is an important information source for investors, it is a voluntary disclosure. There are no formal SEC requirements that require companies to issue an EA,[7] specify the information that should be included in an EA, or require that the information in an EA be certified by an independent audit firm (Bochner and Blake 2008; Steinberg 2009). Consequently, there is large variation in the amount of GAAP financial statement information that management provides in the EA disclosure, ranging from limited summary income statement information to full GAAP financial statements.

[7] There are NYSE and NASDAQ requirements that listed companies provide an EA to the market; however, there are no requirements regarding content and the timing of the EA release or that the information included be audited (Evans 2015).

Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.
The Accounting Review 2016.91:677-705.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

Regulators and practitioners strongly encourage companies to include in the EA more detailed GAAP financial statement disclosures. The National Investor Relations Institute's (NIRI) Standards of Practice on Earnings Release Content urges companies to include in their EA a balance sheet and cash flow statement, in addition to the traditional income statement (NIRI 2012). The Chartered Financial Analyst (CFA) Institute and the SEC Committee on Improvements in Financial Reports (CIFiR) also make similar recommendations (CFA Institute 2007; SEC 2008).

Prior research finds that the information content of the EA has increased over time, due to managers providing more detailed income statement, balance sheet, and cash flow statement information (Francis et al. 2002; Collins et al. 2009; Pawlewicz 2011). More detailed GAAP financial statement information allows investors to better evaluate the quality of earnings for valuation purposes and to form their expectations of a company's future earnings and cash flows (Chen et al. 2002; Baber et al. 2006; Wasley and Wu 2006; D'Souza et al. 2010). Furthermore, providing the full set of financial statements (especially the cash flow statement) increases the saliency of accrual and cash flow information, improving both unsophisticated and sophisticated investor valuations of earnings (Miao et al. 2013). Prior research also finds that companies with increased information demands (i.e., greater analyst following, more institutional investors, and larger market capitalization), noisier earnings, bad news, and higher litigation risk provide more detailed EA GAAP summary disclosures (Chen et al. 2002; D'Souza et al. 2010).

Management faces increased risks of litigation and personal reputation loss if the disclosed EA GAAP summary financial statement information differs from what is subsequently reported in the 10-K filing. The historical GAAP financial statements provided in the EA are subject to the anti-fraud provisions of Section 10(b) and Rule 10b-5 of the Securities Exchange Act of 1934 (Steinberg 2009). Furthermore, research suggests that when litigation risk is high, management provides less disclosure. For example, Cutler et al. (2013) find that companies with greater disclosure during litigation class periods are more likely to face significant settlement costs, and Rogers and Van Buskirk (2009) find that companies reduce disclosure following class action lawsuits. Managers develop a strong reputation for disclosure practices that are credible and reliable, which is rewarded by the market (Trueman 1986; Yang 2012). Conversely, management is penalized for lower quality disclosures with an increased likelihood of termination and lower future job prospects (Desai et al. 2006). Given the litigation and reputation risk associated with issuing EA disclosures that are subject to change, management may rely on mechanisms that increase their confidence in the financial reporting process.

Audit Completeness, Quality, and EA GAAP Disclosures

It is important to determine whether auditing and, more specifically, more complete and higher quality audits, influence a manager's decision to provide more detailed information. Although there is no formal audit requirement for the EA disclosure, the audit may play a role in management's EA disclosure decisions by improving the quality of the company's financial reporting system, the financial statement outputs of that system, and the underlying financial statement balances. In turn, management will have more confidence that the underlying financial statement balances will not change between the EA release date and the 10-K filing date. Consequently, management is less exposed to litigation risk and reputation loss, with more complete and higher quality audits making it less costly to release a fuller set of financial statements.

Waiting until the audit is complete (or substantially complete) before releasing the annual EA reduces the uncertainty surrounding specific account balances and provides management with greater confidence that the underlying financial statement balances will not change between the EA date and the 10-K date. Practitioner organizations strongly encourage companies to actively involve the external auditor in the EA disclosure process and to wait until the audit is complete prior to releasing the EA to the market to reduce concerns that subsequent adjustments will have to be made (Bochner and Blake 2008; Diamond and Yevmenenko 2011; Deloitte 2012). Auditors require considerable time to evaluate the appropriateness of complex accounting estimates that are included in the financial statements (e.g., bad debt reserves, inventory obsolescence reserves, fair value estimates, and income tax provision) (Lambert, Jones, Brazel, and Showalter 2015). Consequently, releasing the EA before the auditor has completed substantive testing increases the likelihood that the financial statement balances included in the EA will differ from the audited numbers in the 10-K filing (Bronson et al. 2011).

However, management also faces the demands for timely and consistent EA disclosures and potential market penalties for deviating from expected disclosure behavior,[8] resulting in pressures to release the EA prior to the completion of the audit. In this situation, management is likely to release less detailed summary financial information as the financial statement balances

[8] Management disclosure practices are typically sticky and consistent with prior years as the market will react negatively to deviations in disclosure behavior (Einhorn and Ziv 2008; Tang 2012). Furthermore, the literature examining EA delays and the "good news early, bad news late" phenomenon provide evidence of market penalties when companies choose to delay an anticipated EA as the market infers that the delay is due to bad news (Kross 1981; Givoly and Palmon 1982; Kross and Schroeder 1984; Begley and Fischer 1998; Bagnoli, Kross, and Watts 2002).

are not final and management has less confidence in the underlying balances. These arguments lead to the first hypothesis, stated in alternative form:

H1: *Ceteris paribus,* companies that release the EA when the audit is more complete provide more detailed EA GAAP financial statement disclosures.

Management's investment in higher quality audits will increase the quality of the financial reporting system and increase the likelihood that significant accounting issues are addressed in a timely and reliable manner at the earnings release date, increasing management's confidence in the financial statement balances. The Big 4 audit firms are international networks of member firms that share personnel resources and expertise across offices and engagement teams (Francis 2004, 2011; Francis and Yu 2009). Furthermore, the Big 4 firms have national consulting offices to assist in addressing complex accounting issues surrounding the financial statements. Consistent with this notion, the extant audit quality literature has found a positive association between auditor size and financial reporting quality in the 10-K (e.g., Dopuch and Simunic 1980; DeAngelo 1981; Becker, DeFond, Jiambalvo, and Subramanyam 1998; DeFond and Zhang 2014). Consequently, management will have more confidence in the financial statement balances increasing the likelihood that they provide more detailed EA GAAP financial statement disclosures.

Companies that demand higher levels of assurance over their financial statement balances are likely to pay higher-than-expected audit fees to ensure that the external auditor can allocate more resources (i.e., personnel and expertise) to ensure a timely and high-quality audit. Audit firms seek to maximize engagement profits while providing audits that are in compliance with auditing standards and reducing exposure to future litigation losses (Simunic 1980). Consequently, audit firms that receive lower-than-expected audit fees must either provide the necessary level of assurance at a loss or reduce the level of assurance to remain profitable. Therefore, companies that value higher quality audits will pay excess audit fees to increase the quality of their financial statement information (Blankley et al. 2012; Ball et al. 2012).[9] This argument leads to the second hypothesis, stated in alternative form:

H2: *Ceteris paribus,* companies that receive higher quality audits (measured in terms of audit firm size and excess audit fees) provide more detailed EA GAAP financial statement disclosures.

III. RESEARCH DESIGN AND SAMPLE SELECTION

Research Design

I estimate the following ordinary least squares (OLS) regression model to test the predictions that more complete audits (H1) and higher quality audits (H2) are positively associated with companies providing more detailed GAAP financial statement disclosures in their annual EA:

$$\begin{aligned} DR_{i,t} = {} & \beta_0 + \beta_1 AUDCOMP_{i,t} + \beta_2 BIGN_{i,t} + \beta_3 ABFEES_{i,t} + \beta_4 EARVZ_{i,t} + \beta_5 EA_LAG_{i,t} + \beta_6 SUPDISC_{i,t} \\ & + \beta_7 INFOENVT_{i,t} + \beta_8 LITRISK_{i,t} + \beta_9 BADNEWS_{i,t} + \beta_{10} COMPLX_{i,t} + \beta_{11} DFACT_{i,t} + \beta_{12} LNASSETS_{i,t} \\ & + \beta_{13} ACCEL_{i,t} + \beta_{14} LGACCEL_{i,t} + industry\ fixed\ effects + year\ fixed\ effects + \varepsilon_{i,t} \end{aligned} \quad (1)$$

The model is adapted from D'Souza et al. (2010); variables are defined in Appendix A and discussed further below.

Measure of EA GAAP Disclosure Detail (Dependent Variable)

I measure EA GAAP financial statement disclosure details using the total GAAP financial statement disclosure ratio (DR_{TOT}) from D'Souza et al. (2010),[10] which is calculated as follows:

$$\frac{\text{Number of Financial Statement Line Items Disclosed in the Earnings Announcement}}{\text{Number of Financial Statement Line Items Disclosed in the Initial 10-K Filing}} \quad (2)$$

[9] I use abnormal audit fees (actual fee minus expected fee, obtained from a standard audit fee model) to determine whether audit fees are above or below expectations, with further details discussed in Section III. The literature presents two theories as to what positive versus negative abnormal fees represent. Some studies argue that positive abnormal fees represent higher fees that create an economic bond between the audit firm and the client, resulting in reduced auditor independence and audit quality (Causholli, DeMartinis, Hay, and Knechel 2010; Kinney and Libby 2002; Choi, Kim, and Zhang 2010). Other studies posit that positive abnormal audit fees indicate greater auditor effort and/or resources committed to auditor verification, resulting in higher quality audits (Causholli et al. 2010; Blankley et al. 2012; Ball et al. 2012). Consistent with the later stream of research, I use positive abnormal audit fees to represent increased audit quality.

[10] Evans (2015) uses the D'Souza et al. (2010) balance sheet ratio to examine the association between disclosure commitment and cost of capital providing additional validation of this measure.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

This ratio is based on the fourth-quarter EA.[11] The numerator represents the number of non-missing data items from Compustat's Quarterly Preliminary History database, which is from companies' quarterly EAs. The denominator represents the number of non-missing data items from Compustat's Unrestated Quarterly database, which is from the initial SEC periodic filing. The ratios are based on a total of 84 data items from the three financial statements (23 from the income statement, 28 from the balance sheet, and 33 from the cash flow statement), with specific definitions provided in Appendix B. I further examine three separate ratios for the income statement (DR_{IS}), balance sheet (DR_{BS}), and cash flow statement (DR_{CF}) to provide further insights into which financial statement disclosures drive the overall total disclosure ratio result.

Measure of Audit Completeness

Consistent with AU 530 (adopted by the PCAOB), I deem the audit report date as the date of audit completion.[12] The measure of audit completeness (*AUDCOMP*) is calculated in two parts based on whether the audit is complete at the EA date. When the EA is released on or after the audit report date, *AUDCOMP* is set equal to 0 (i.e., the audit is complete at the EA date), and when the EA is released before the audit report date, *AUDCOMP* equals the number of days between the EA date and the audit report date, resulting in negative values.[13] *AUDCOMP* captures the degree of audit completeness at the EA date, and it assumes that each day closer to the audit report date is incrementally more complete. Consistent with H1, I expect a positive coefficient on *AUDCOMP*.

Measures of Audit Quality

I use two measures for *audit quality, BIGN* (i.e., *size/resources*) and *ABFEES* (i.e., *excess audit fees*). *BIGN* is an indicator variable that captures whether the company engages one of the Big 4 firms. *ABFEES* is the residual from a standard audit fee model modified from prior studies (e.g., Hay, Knechel, and Wong 2006; Hogan and Wilkins 2008), scaled by the natural log of total audit fees to adjust for size effects, with positive (negative) values indicating that the audit fees are higher (lower) than model expectations (see Appendix A for the specific model). I make the assumption that the residual from this model captures increased auditor resources, which facilitates greater audit effort and timeliness (Causholli et al. 2010; Ball et al. 2012; Blankley et al. 2012). Consistent with the prediction of H2, I expect positive coefficients on *BIGN* and *ABFEES*.

Control Variables for Other Disclosure Incentives

Bronson et al. (2011) find that companies issuing their EA before the audit report date are more likely to experience an EA revision where net income reported in the EA differs from what is reported in the subsequent 10-K filing. I include a dichotomous variable that captures whether a company experiences an EA revision (*EARVZ*) during the current period and expect a negative coefficient. Prior literature documents that earlier EA releases are associated with larger companies, greater information demand, good news, more sophisticated accounting systems, and more detailed GAAP disclosures (Kross 1981; Givoly and Palmon 1982; Sengupta 2004; D'Souza et al. 2010). Therefore, it is important to hold constant many of the key risks/determinants associated with the timing of the earnings release so that the coefficient on *AUDCOMP* fully captures the impact of more complete audits on management GAAP disclosure decisions. I include a control measure that is the number of days from the fiscal year-end to the EA release date (*EA_LAG*).[14]

Chen et al. (2002) find that companies are more likely to provide supplemental balance sheet disclosures when net income is a poor predictor of future earnings and cash flows. I include a composite measure (*SUPDISC*) from D'Souza et al. (2010),

[11] D'Souza et al. (2010) verify the accuracy of the quarterly Preliminary History and Unrestated Quarterly files and indicate that the annual files are less reliable due to Compustat's data-collection practices. The Q4 disclosure ratio is a good proxy for the annual disclosure ratio because EAs typically provide both quarterly and annual information side by side for the income statement and cash flow statements and are a point in time for the balance sheet.

[12] AU 530 (available at: http://www.pcaobus.org/standards/auditing/pages/au530.aspx), paragraph 1, states, "[T]he auditor should date the audit report no earlier than the date on which the auditor has obtained sufficient appropriate evidence to support the auditor's opinion." Prior studies also deem the audit report date to be the date of audit completion (e.g., Bronson et al. 2011).

[13] I also use an alternative measure of *AUDCOMP* that is not truncated at 0 (i.e., instances where the EA date is after the audit report date would be assigned positive values). I find results consistent with Tables 5 and 6.

[14] I perform two additional sensitivity tests to control for the timing of the EA release rather than including *EA_LAG* in the model. First, I separate the main sample into two subsamples: observations with complete (n = 6,278) and incomplete audits (n = 16,000). I then perform a one-to-one match based on *EA_LAG,* two-digit SIC code, and fiscal year, resulting in a sample of 9,244 observations (4,622 matched pairs). This process results in treatment (i.e., complete audits) and control (i.e., incomplete audits) observations having similar EA release dates (mean of 46.5 versus 47.1 days). In untabulated results, the coefficient on *AUDCOMP* continues to be positive and significant (Coeff. = 0.003; t-stat = 2.06; $p < 0.05$). The second analysis replaces *EA_LAG* with a measure of unexpected *EA Lag*. I calculate unexpected *EA Lag* as the residual from a model that regresses *EA_LAG* on the determinants included in Model (3), as well as prior-year *EA_LAG*. In untabulated results, the coefficient on *AUDCOMP* continues to be positive and significant (Coeff. = 0.0005; t-stat = 2.93; $p < 0.01$).

American Accounting Association

based on six attributes documented in Chen et al. (2002), to capture the demand for supplemental GAAP disclosures. I expect a positive coefficient on *SUPDISC*.

The voluntary disclosure literature has identified four economic constructs that are likely to impact the disclosure strategies of companies: the richness of the information environment, litigation risk, proprietary costs, and bad news (Healy and Palepu 2001). Consistent with D'Souza et al. (2010), I use factor analysis to reduce the dimensionality of 16 specific variables used in the literature to proxy for the above disclosure incentives; further details can be found in Appendix C.[15] The factor analysis results in five factors. The first factor captures the richness of the company's information environment (*INFOENVT*). I predict a positive coefficient on *INFOENVT,* which is consistent with management providing more detailed EA disclosures when there is greater stakeholder demand (Diamond 1985; Lang and Lundholm 1993; Scott 1994; Botosan 1997; Frankel, Johnson, and Skinner 1999; Healy and Palepu 2001; Verrecchia 2001; Botosan and Plumlee 2002; Chen et al. 2002; Bushee, Matsumoto, and Miller 2002; Graham, Harvey, and Rajgopal 2005).

The second factor captures the company's exposure to litigation risk (*LITRISK*). I do not make a prediction for the sign of the coefficient since prior literature has demonstrated mixed results: some studies suggest that litigation risk results in more detailed disclosures (Botosan and Harris 2000; Bushee et al. 2002; Chen et al. 2002; Field, Lowry, and Shu 2005), while others suggest that it results in less detailed disclosures (Rogers and Van Buskirk 2009; Cutler et al. 2013). The third factor captures the existence of bad news (*BADNEWS*). Chen et al. (2002) suggest that companies with less informative earnings (e.g., bad news information) provide balance sheet disclosures as a context for the bad news, suggesting a positive coefficient. However, Wasley and Wu (2006) suggest that companies are less likely to disclose bad news cash flow information. Accordingly, I do not make a prediction regarding the coefficient on *BADNEWS*.

The fourth factor captures operational complexity (*COMPLX*) and loads positively on market value of equity, business segments, and total shareholders. I expect that investors demand more detailed EA GAAP disclosures with increased operational complexity. The final factor (*DFACT*) loads positively on special items and negatively on market-to-book ratio. I do not make predictions regarding the coefficient sign on *DFACT*.

I include the natural log of assets (*LNASSETS*) to control for company size.[16] I also include indicator variables capturing filer status (*ACCEL* and *LGACCEL*) to control for differential SEC filing and disclosure demands. Industry and year fixed effects are included to control for industry-wide and year-specific macroeconomic effects.

Control for Endogeneity

The factors that influence a company's timing of the earnings release in relation to audit completion (i.e., complete versus incomplete) and/or audit firm choice (i.e., Big N versus non-Big N) might also be associated with the decision to disclose more or less GAAP information. To partly mitigate this concern, I use two approaches: (1) propensity score matching (PSM) to construct two samples, and (2) a first-differencing (i.e., change model) research design.

PSM is beneficial because it can achieve covariate balance on observable characteristics between treatment and control groups (i.e., differences in means are not statistically significant), allowing for better specification and causal inference (Rosenbaum and Rubin 1983). The first PSM test estimates the probability that a company releases the EA on or after the audit report date using the following probit model from Bronson et al. (2015), with variable definitions found in Appendix A:

$$\begin{aligned} AUDEA = {} & \beta_0 + \beta_1 LNMVE_{i,t} + \beta_2 ANALYST_FOLL_{i,t} + \beta_3 ANALYST_COVG_{i,t} + \beta_4 INSTHOLD_{i,t} + \beta_5 ROA_{i,t} + \beta_6 OCF_{i,t} \\ & + \beta_7 LEV_{i,t} + \beta_8 UE_NEG_{i,t} + \beta_9 LOSS_{i,t} + \beta_{10} MBR_{i,t} + \beta_{11} MKTCONC_{i,t} + \beta_{12} LIT_{i,t} + \beta_{13} SPECITM_{i,t} \\ & + \beta_{14} LNBSEG_{i,t} + \beta_{15} INVREC_{i,t} + \beta_{16} MERGER_{i,t} + \beta_{17} GC_{i,t} + \beta_{18} BIGN_{i,t} + \beta_{19} YE_{i,t} + \beta_{20} ACCEL_{i,t} \\ & + \beta_{21} LGACCEL_{i,t} + year\ and\ industry\ controls + \varepsilon_{i,t} \end{aligned} \tag{3}$$

The sample is based on a one-to-one match, without replacement, of treatment observations (EA on or after the audit report date) to control observations (EA issued before audit report date) within a caliper range of 3 percent (Guo and Fraser 2010). This results in 3,754 matched pairs (7,508 observations). Covariate balance is achieved on all explanatory variables with the exception of *BIGN,* which is included in the second-stage model.[17]

[15] Factor analysis provides more concrete measures of the economic constructs explored in the voluntary disclosure literature, as opposed to including 16 different measures that may proxy for multiple constructs and introduce multicollinearity into the model. I reestimate Model (1) replacing the factors with the 16 detailed variables, and results are qualitatively similar to what is reported later in Tables 5 and 6.

[16] As an additional test, I reestimate the multivariate models by SEC filer status group (i.e., large accelerated filers, accelerated filers, and non-accelerated filers). The coefficients on *AUDCOMP* in all three specifications are positive and significant, consistent with the predictions of H1 and results presented in Table 5.

[17] Untabulated *BIGN* means for the treatment and control groups are 0.807 and 0.827, respectively, with a mean difference t-test of −2.27 ($p = 0.023$).

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

TABLE 1

Sample Selection

Cross merge of Compustat NA (U.S. registrants) and Audit Analytics Opinion File for 2001 to 2011	66,705
Less: Observations with SIC codes between 6000 and 6999	(14,433)
Less: Missing EA date, EA released before year-end, or EA released after SEC filing deadline	(1,285)
Less: Delayed 10-K filings (i.e. filed greater than 105 days per Audit Analytics)	(8,336)
Less: EA filed concurrently with or was 10-K filing	(12,015)
Observations available for analysis of EA trends	30,636
Less: Missing CRSP and Compustat data to create variables	(7,502)
Sample used to evaluate influential observations (see footnote 22)	23,134
Less: Influential outliers with R-Student values greater than 2 in absolute value	(856)
Sample used for 2001 to 2011 multivariate analyses	22,278
Observations for audit completeness PSM sample	7,508
Observations for audit quality PSM sample	5,762
Observations for change model (i.e. first-differencing)	19,115

The second PSM test estimates the probability that a company engages a non-Big N audit firm using the following model modified from Lawrence, Minutti-Meza, and Zhang (2011), with variable definitions found in Appendix A:

$$NONBIGN = \beta_0 + \beta_1 LNASSETS_{i,t} + \beta_2 ATURN_{i,t} + \beta_3 CURR_{i,t} + \beta_4 LEV_{i,t} + \beta_5 ROA_{i,t} + year\ and\ industry\ controls + \varepsilon_{i,t} \quad (4)$$

The sample is based on a one-to-one match, without replacement, of treatment observations (non-Big 4 client) and control observations (Big 4 client) within a caliper range of 3 percent. This results in 2,881 matched pairs (5,762 observations). Covariate balance is achieved on all explanatory variables.

Finally, I estimate Model (1) using a change specification. First-differencing is appealing, as it controls for any time-invariant, unobservable company characteristics that potentially explain a company's disclosure decision and audit quality (Wooldridge 2002).

Sample Selection

Table 1 provides a summary of the sample selection. I begin with 66,705 observations from years 2001 to 2011 that represent the intersection of Compustat North America (U.S. registrants only) and Audit Analytics. I eliminate 14,433 financial institutions (SIC codes between 6000 and 6999) due to the industry's regulatory disclosure requirements, and 1,285 observations with missing EA dates. I then eliminate 8,336 observations in which the 10-K filing was significantly delayed (i.e., 10-K filed after the possible 15 day NT 10-K extension), and 12,015 observations where the EA is concurrent with the 10-K filing date, as the focus of this study is on companies that elect to issue an EA prior to the 10-K filing.[18] This results in a final sample of 30,636 observations available for the disclosure trend analysis in Table 2. I then lose 7,502 observations due to missing values in either Compustat or CRSP necessary to construct the control variables. I then eliminate 856 influential outlier observations.[19] This results in a final sample of 22,278 observations.

[18] Companies that release earnings concurrent with the 10-K differ on overall disclosure strategies, market information demands, accounting sophistication, and investments in audit quality. As a sensitivity test, I include these observations (27,864 observations with available data to create control variables). The disclosure ratios for these observations are coded 1 as the full GAAP financial statements are available in the 10-K filing. The results (untabulated) are consistent with Tables 5 and 6 with the exception of the coefficient on *BIGN,* which is no longer significant in the total GAAP and cash flow statement ratio models.

[19] To limit the influence of outliers, I exclude influential outlier observations with R-student values greater than or equal to 2 (Belsley, Kuh, and Welsch 1980), resulting in a loss of 856 (1,058) observations of the 23,134 (20,173) available observations. Without the R-student restriction, results (untabulated) are consistent with what is subsequently reported in Table 5 for the full sample and change model.

American Accounting Association

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

TABLE 2

Descriptive Earnings Announcement Disclosure Trends

Panel A: Earnings Announcement Disclosure Trends in Relation to Audit Completion

	2001		2002		2003		2004		2005		2006	
	Count	%	Count	%	Count	%	Count	%	Count	%	Count	%
EA on or after audit report date	2,430	72%	2,343	71%	1,859	60%	923	32%	562	20%	429	15%
EA issued before audit report date	968	29%	956	29%	1,230	40%	1,996	68%	2,314	80%	2,363	85%
Total	3,398		3,299		3,089		2,919		2,876		2,792	

Panel B: Earnings Announcement Disclosure Trends in Relation to Audit Completion, continued

	2007		2008		2009		2010		2011	
	Count	%	Count	%	Count	%	Count	%	Count	%
EA on or after audit report date	314	12%	298	12%	167	7%	116	5%	106	5%
EA issued before audit report date	2,354	88%	2,222	88%	2,289	93%	2,235	95%	2,162	95%
Total	2,668		2,520		2,456		2,351		2,268	

Panel C: Earnings Announcement Disclosure Trends in Relation to Audit Completion, continued

	2001		2002		2003		2004		2005		2006	
	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.
EA lag	43.6	42.0	44.2	42.0	43.7	42.0	45.6	44.0	47.4	46.0	48.2	46.0
Audit report date lag	46.4	43.0	47.7	43.0	49.0	48.0	60.4	66.0	64.6	67.0	65.2	66.0
10-K file lag	83.1	87.0	81.8	86.0	76.1	75.0	76.1	75.0	72.4	74.0	69.9	72.0
Diff. between EA and audit report lags	−2.8	1.0	−3.5	1.0	−5.4	0	−14.8	−15.0	−17.2	−18.0	−17.0	−17.5

Panel D: Earnings Announcement Disclosure Trends in Relation to Audit Completion, continued

	2007		2008		2009		2010		2011	
	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.
EA lag	48.5	46.0	48.0	47.0	47.8	47.0	46.8	46.0	47.0	46.0
Audit report date lag	64.9	60.0	63.6	61.0	63.6	60.0	62.8	60.0	62.6	60.0
10-K file lag	67.4	65.0	65.6	61.0	64.3	60.0	63.2	60.0	62.9	60.0
Diff. between EA and audit report lags	−16.3	−16.0	−15.7	−15.0	−15.8	−15.0	−15.9	−15.0	−15.6	−14.0

Panel E: Earnings Announcement Disclosure Ratio Trends

	2001		2002		2003		2004		2005		2006	
	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.
DR_{TOT}	0.42	0.45	0.46	0.47	0.51	0.50	0.54	0.52	0.55	0.52	0.56	0.53
DR_{IS}	0.84	0.86	0.84	0.86	0.85	0.86	0.86	0.87	0.87	0.87	0.87	0.87
DR_{BS}	0.52	0.61	0.57	0.68	0.61	0.75	0.64	0.78	0.65	0.77	0.66	0.79
DR_{CF}	0.04	0.00	0.09	0.00	0.17	0.00	0.22	0.00	0.24	0.00	0.25	0.00

(continued on next page)

IV. RESULTS

Descriptive Statistics

Earnings Announcement Disclosure Trends 2001 to 2011

Table 2 provides descriptive trends by year for companies that release an EA prior to the 10-K filing. Panels A–D show whether the EA was released before or after the audit report date, EA release lags, audit report date lags, and the number of days

TABLE 2 (continued)

Panel F: Earnings Announcement Disclosure Ratio Trends, continued

	2007		2008		2009		2010		2011	
	Mean	**Med.**	**Mean**	**Med.**	**Mean**	**Med.**	**Mean**	**Med.**	**Mean**	**Med.**
DR_{TOT}	0.58	0.54	0.61	0.55	0.63	0.56	0.64	0.57	0.65	0.58
DR_{IS}	0.88	0.87	0.88	0.87	0.88	0.87	0.89	0.87	0.89	0.91
DR_{BS}	0.68	0.79	0.70	0.79	0.71	0.81	0.72	0.82	0.74	0.82
DR_{CF}	0.28	0.00	0.33	0.00	0.36	0.03	0.39	0.05	0.40	0.06

The above table provides descriptive earnings announcement disclosure trends separated by year. Panels A–D are based on the broader sample of 30,636 company-year observations (see Table 1) to understand the trend of earnings announcement release dates in proximity to the completion of the year-end audit. Panels E and F are based on the sample of 22,278 observations available to perform the multivariate analyses and provide the disclosure ratio trends over time.

between the EA release and the audit report date. The percentage of companies that release annual earnings before the completion of the audit is increasing over time, from 29 percent in 2001 to 95 percent in 2011, with a significant shift occurring from 2003 to 2004 (40 percent versus 68 percent), the first year of the Sarbanes-Oxley Act (SOX) Section 404(b) integrated audits. This shift is due to management maintaining their historical earnings release date (average of 44 days for 2001 to 2003 versus 47 days for 2004 to 2011), while audits take, on average, 16 days longer during the post-AS2/3 period (average audit lag of 48 days for 2001 to 2003 versus 64 days for 2004 to 2011).[20] The 10-K filing lags have decreased from a high of 83 days in 2001 to a low of 63 days in 2011 due to the change in SEC filing deadlines for large accelerated and accelerated filers (Doyle and Magilke 2013).

Panels E and F of Table 2 present the disclosure ratio trends from 2001 to 2011 for the sample used in the multivariate analysis (22,278 company-year observations). Total GAAP financial statement disclosures have increased from an average of 42 percent in 2001 to 65 percent in 2011. This is largely due to companies providing more detailed balance sheet disclosures (52 percent in 2001 versus 74 percent in 2011) and cash flow statement disclosures (4 percent in 2001 versus 40 percent in 2011). Income statement disclosures remain consistent over time (84 percent in 2001 versus 89 percent in 2011).

Sample Statistics

Table 3 provides descriptive statistics for the variables used in the multivariate analyses. The total disclosure ratio (DR_{TOT}) has considerable variability (SD = 0.223), largely due to the balance sheet ratio (SD = 0.312) and cash flow statement ratio (SD = 0.393). The income statement ratio has less variability (SD = 0.091), which is consistent with most companies providing detailed income statement disclosures (Francis et al. 2002). These statistics are consistent with prior evidence that differences in EA GAAP disclosure details derive mainly from supplemental balance sheet and cash flow statement information (Chen et al. 2002; Collins et al. 2009; D'Souza et al. 2010). Sample observations, on average, issue EAs 15 days (median of 12 days) before the audit report date, with 28.2 percent of the sample waiting until on or after the audit report date to release earnings. Consistent with prior studies, 83.7 percent of the sample is audited by a Big 4 firm. The mean (median) value of *ABFEES* is −0.001 (0.000), consistent with the variable design (i.e., residual from standard audit fee model). The remaining control variables have mean and median values and distributions consistent with prior disclosure studies (e.g., D'Souza et al. 2010).

Table 4 provides the Pearson correlations for the dependent, independent, and control variables. The measure of audit completeness (*AUDCOMP*) is negatively correlated with the disclosure ratios, which is inconsistent with the prediction of H1. This is largely due to smaller companies disclosing less GAAP information (correlations between *LNASSETS* and disclosure ratios are positive) and also being more likely to wait for audit completion prior to releasing earnings (correlation between *LNASSETS* and *AUDCOMP* is negative). Thus, it is important to fully control for company size in order to capture the effects of audit completeness. Consistent with the prediction of H2, the correlation between the audit quality measures (*BIGN* and *ABFEES*) and the disclosure ratios is positive and significant.

[20] PCAOB Auditing Standard No. 5 (AS5) superseded AS2 effective November 15, 2007, scaling the requirements necessary to conduct a Section 404(b) internal control audit (PCAOB 2007). As evidenced in Figure 1 and Table 2, AS5 did not result in more timely audits in relation to the annual earnings release.

The Accounting Review 2016;91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

American Accounting Association

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

TABLE 3

Descriptive Statistics Main Sample

Panel A: Descriptive Statistics for Levels Regression (n = 22,278)

Variables	Mean	25%	Median	75%	SD
DR_{TOT}	0.555	0.400	0.525	0.723	0.223
DR_{IS}	0.867	0.826	0.870	0.952	0.091
DR_{BS}	0.651	0.464	0.786	0.926	0.312
DR_{CF}	0.245	0.000	0.000	0.633	0.393
AUDCOMP	−14.803	−26.000	−12.000	0.000	14.539
AUDEA	0.282	0.000	0.000	1.000	0.450
BIGN	0.837	1.000	1.000	1.000	0.369
ABFEES	−0.001	−0.025	0.000	0.024	0.038
EARVZ	0.039	0.000	0.000	0.000	0.193
EA_LAG	45.269	33.000	44.000	56.000	15.982
SUPDISC	2.772	2.000	3.000	3.000	1.035
INFOENVT	0.024	−0.679	0.217	0.788	1.006
LITRISK	0.032	−0.678	−0.042	0.701	1.002
BADNEWS	−0.008	−0.670	−0.328	0.575	1.007
COMPLX	−0.031	−0.720	−0.158	0.563	0.974
DFACT	−0.008	−0.656	0.034	0.635	1.004
LNASSETS	5.994	4.653	5.911	7.249	1.870
ACCEL	0.417	0.000	0.000	1.000	0.493
LGACCEL	0.398	0.000	0.000	1.000	0.490

Panel B: Descriptive Statistics for Change Regression (n = 19,115)

Variables	Mean	25%	Median	75%	SD
ΔDR_{TOT}	0.007	−0.013	0.000	0.019	0.053
ΔDR_{IS}	0.003	0.000	0.000	0.004	0.067
ΔDR_{BS}	0.011	0.000	0.000	0.015	0.114
ΔDR_{CF}	0.005	0.000	0.000	0.000	0.074
Δ*AUDCOMP*	−1.279	−4.000	0.000	2.000	13.296
Δ*AUDEA*	−0.070	0.000	0.000	0.000	0.424
Δ*BIGN*	−0.012	0.000	0.000	0.000	0.131
Δ*ABFEES*	0.002	−0.011	0.001	0.016	0.026
Δ*EARVZ*	0.003	0.000	0.000	0.000	0.247
Δ*EA_LAG*	0.838	0.000	0.000	4.000	7.427
Δ*SUPDISC*	−0.030	−1.000	0.000	0.000	0.891
Δ*INFOENVT*	0.072	−0.068	0.041	0.163	0.326
Δ*LITRISK*	0.014	−0.223	0.014	0.247	0.427
Δ*BADNEWS*	−0.063	−0.445	−0.035	0.367	0.728
Δ*COMPLX*	0.020	−0.159	0.024	0.203	0.358
Δ*DFACT*	−0.003	−0.553	−0.018	0.523	1.123
Δ*LNASSETS*	0.057	−0.043	0.049	0.150	0.275
Δ*ACCEL*	−0.011	0.000	0.000	0.000	0.395
Δ*LGACCEL*	0.010	0.000	0.000	0.000	0.287

The above table provides descriptive statistics for the dependent, independent, and control variables used in the levels main sample analysis and change analysis. The change variables presented in Panel B are the difference between current-year and prior-year values for the respective variables. All variables are defined in Appendix A.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

TABLE 4

Correlation Matrix for Main Analysis (n = 22,278)

Panel A: Correlation Variables DR_{TOT} to *EARVZ*

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
1. DR_{TOT}									
2. DR_{IS}	0.447								
3. DR_{BS}	0.749	0.271							
4. DR_{CF}	0.876	0.306	0.353						
5. *AUDCOMP*	−0.132	−0.075	−0.111	−0.103					
6. *AUDEA*	−0.211	−0.153	−0.126	−0.195	0.638				
7. *BIGN*	0.118	0.108	0.047	0.120	−0.081	−0.042			
8. *ABFEES*	0.077	0.025	0.073	0.058	**0.002**	−0.017	**−0.011**		
9. *EARVZ*	**−0.011**	**−0.010**	**−0.004**	**−0.012**	−0.156	−0.081	0.017	0.039	
10. *EA_LAG*	−0.199	−0.073	−0.149	−0.174	0.467	0.137	−0.317	**0.015**	−0.059
11. *SUPDISC*	**−0.001**	**0.004**	0.069	−0.055	−0.048	**−0.006**	0.063	0.018	0.038
12. *INFOENVT*	0.301	0.209	0.153	0.299	−0.193	−0.286	0.318	**−0.005**	**0.000**
13. *LITRISK*	0.074	0.029	0.137	**0.004**	−0.088	−0.017	0.071	0.038	0.036
14. *BADNEWS*	−0.151	−0.067	−0.028	−0.194	0.080	0.104	−0.064	0.041	0.038
15. *COMPLX*	0.141	0.026	−0.016	0.212	−0.109	−0.097	0.220	0.052	**0.011**
16. *DFACT2*	0.034	0.038	0.038	0.015	**0.004**	−0.026	0.033	**0.005**	0.047
17. *LNASSETS*	0.262	0.174	0.037	0.330	−0.164	−0.231	0.382	0.036	**0.007**
18. *ACCEL*	−0.057	0.020	0.027	−0.107	0.021	**0.011**	−0.020	−0.041	**0.013**
19. *LGACCEL*	0.229	0.115	0.052	0.280	−0.134	−0.170	0.294	0.033	**−0.010**

Panel B: Correlation Variables *SUPDISC* to *LGACCEL*

	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
11. *SUPDISC*	**−0.011**								
12. *INFOENVT*	−0.397	−0.073							
13. *LITRISK*	−0.167	0.248	0.032						
14. *BADNEWS*	0.219	0.384	−0.326	0.165					
15. *COMPLX*	−0.339	−0.111	0.292	0.068	−0.272				
16. *DFACT2*	0.020	0.072	**−0.009**	−0.014	0.178	−0.102			
17. *LNASSETS*	−0.445	−0.117	0.655	−0.007	−0.397	0.667	0.071		
18. *ACCEL*	0.118	0.059	−0.105	−0.070	0.136	−0.286	0.020	−0.278	
19. *LGACCEL*	−0.403	−0.068	0.566	0.102	−0.358	0.563	−0.080	0.729	−0.688

The above table provides Pearson correlations, with all non-bold values being significant at the 0.05 level (two-tailed p-values).
Variables are defined in Appendix A.

Multivariate Results

I begin by examining the effect of audit completeness and quality on overall GAAP financial disclosures (DR_{TOT}). Table 5, Column (1) presents the multivariate result for the full sample.[21] Consistent with the prediction of H1, the coefficient on the audit completeness measure is positive and significant (*AUDCOMP* $p < 0.01$).[22] Consistent with the prediction of H2, the coefficients on the audit quality measures are positive and significant (*BIGN* and *ABFEES* $p < 0.01$). The results suggest that more complete audits and higher quality audits (i.e., engaging a larger audit firm and paying excess fees) are associated with management providing more detailed EA GAAP disclosures. These results are also economically significant, as an increase in the inter-quartile range of the audit completeness and quality proxies increases total GAAP disclosures by 14.41 percent.

[21] I test each regression model for multicollinearity and find that none of the variance inflation factors (VIFs) exceed 10 (highest value is 6.49 for *LGACCEL*).

[22] Including control variables for company size (*LNASSETS*), information environment (*INFOENVT*), and timing of the earnings release (*EA_LAG*) in the multivariate model contributes to the positive coefficient on *AUDCOMP,* supporting H1.

American Accounting Association

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

TABLE 5

Multivariate Results for Total GAAP Disclosures (DR_{TOT})

	Pred. Sign	Full Sample (1) Coeff. (t-stat)	PSM Sample (Audit Comp.) (2) Coeff. (t-stat)	PSM Sample (Audit Quality) (3) Coeff. (t-stat)	Change Model (see note below) (4) Coeff. (t-stat)
AUDCOMP	+	0.0009***	0.0005***	0.0010***	0.0001***
		(5.96)	(3.40)	(3.81)	(3.15)
BIGN	+	0.0394***	0.0414***	0.0234***	0.0066**
		(5.42)	(4.78)	(2.91)	(2.51)
ABFEES	+	0.3503***	0.4238***	0.3575***	0.0340*
		(5.44)	(6.33)	(3.74)	(1.96)
EARVZ	−	−0.0290***	−0.0366***	−0.0303**	−0.0034**
		(−3.91)	(−3.24)	(−2.22)	(−1.98)
EA_LAG	−	−0.0017***	−0.0016***	−0.0018***	−0.0001
		(−8.65)	(−7.43)	(−6.39)	(−0.78)
SUPDISC	+	0.0064***	0.0069**	0.0127***	−0.0008
		(2.70)	(2.34)	(3.61)	(−1.63)
INFOENVT	+	0.0171***	0.0182***	0.0308***	0.0008
		(4.43)	(4.35)	(4.88)	(0.58)
LITRISK	±	0.0088***	0.0086**	0.0165***	0.0006
		(2.62)	(2.31)	(3.04)	(0.56)
BADNEWS	±	−0.0015	0.0024	0.0022	−0.0002
		(−0.69)	(0.83)	(0.70)	(−0.29)
COMPLX	±	0.0021	−0.0003	−0.0058	0.0011
		(0.51)	(−0.06)	(−0.95)	(0.86)
DFACT	±	0.0055***	0.0078***	0.0078***	−0.0005
		(2.71)	(2.90)	(2.79)	(−1.34)
LNASSETS	+	0.0040	0.0037	0.0046	0.0013
		(1.22)	(1.04)	(0.84)	(0.80)
ACCEL	+	0.0269***	0.0281***	0.0237**	−0.0020
		(4.01)	(3.62)	(2.51)	(−1.23)
LGACCEL	+	0.0479***	0.0457***	0.0385**	−0.0019
		(4.38)	(3.53)	(2.12)	(−0.82)
Intercept		0.6293***	0.6224***	0.6262***	0.0027***
		(26.43)	(18.65)	(16.84)	(2.80)
Year FE		Yes	Yes	Yes	Yes
Industry FE		Yes	Yes	Yes	Yes
Observations		22,278	7,508	5,762	19,115
Adjusted R^2		0.282	0.219	0.298	0.015

*, **, *** Indicate $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively, two-tailed tests.

Standard errors are clustered by company to compute t-statistics. The sample period includes years 2001 to 2011. Column (1) presents the result of the full sample; Column (2) presents the result of the audit completeness (i.e., EA released on or after the audit report date versus prior to audit report date) propensity score match sample; and Column (3) presents the result of the audit quality (i.e., Big N versus non-Big N) propensity score match sample. Column (4) presents the results of the change model (i.e., first-differencing), where the dependent, independent, and control variables are transformed to be the difference between the values in years t and $t-1$.

Variable definitions can be found in Appendix A.

Table 5, Columns (2) to (4) present results of the audit completeness PSM sample,[23] audit quality PSM sample, and change specification, respectively. Coefficients on the audit completeness and quality measures in the three specifications remain positive and significant, supporting the predictions of H1 and H2. This reduces concerns about endogeneity impacting the full sample results.

[23] As a sensitivity test, I replace *AUDCOMP* with *AUDEA* for the audit completeness PSM sample. Results are consistent with the results reported in Column (2) of Table 5 and Panel B of Table 6.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

I next examine income statement (DR_{IS}), balance sheet (DR_{BS}), and cash flow (DR_{CF}) disclosures separately to provide greater perspective on the overall GAAP disclosure results. Table 6, Panel A presents the full sample results for the income statement, balance sheet, and cash flow statement ratios. Consistent with H1, the coefficient on audit completeness is positive and significant (*AUDCOMP* $p < 0.01$) for the balance sheet and the cash flow statement ratio models.[24] Consistent with H2, the coefficients on the audit quality measures are positive and significant for all three financial statement ratio models (*BIGN* $p < 0.01$ for all three models; *ABFEES* $p < 0.10$ for DR_{IS}; and $p < 0.01$ for DR_{BS} and DR_{CF}). The results are also economically significant, as increases in the inter-quartile range of the audit completeness and quality proxies increase balance sheet and cash flow statement ratios by 14.41 and 43.36 percent, respectively.

Table 6, Panels B, C, and D present the audit completeness PSM sample, audit quality PSM sample, and first-differencing results, respectively. Results for the audit completeness PSM sample in Panel B are consistent with the full sample results reported in Panel A. Results for the audit quality PSM sample in Panel C are consistent with the full sample results with the exception of the coefficient on *BIGN,* which is no longer significant in the cash flow statement ratio model. Results for the change model in Panel D indicate that year-on-year changes in audit completeness and quality have no effect on changes in income statement disclosures. The coefficient on the audit completeness measure is positive and significant in the balance sheet and cash flow statement models (*AUDCOMP* $p < 0.10$ and $p < 0.01$, respectively), which is consistent with the full sample results. Regarding audit quality, the coefficient on the auditor size measure is positive and significant in the balance sheet ratio model (*BIGN* $p < 0.05$), but not significant in the cash flow model. The coefficient on the excess audit fee measure is positive and significant in the cash flow statement ratio model (*ABFEES* $p < 0.05$), but not significant in the balance sheet model.

Additional Analyses

Q4/Annual versus Quarterly EA GAAP Disclosures

To strengthen the causal inferences, I perform a falsification test comparing Q4/annual EA releases (subject to audit requirements) to Q1, Q2, and Q3 EA releases (subject to review requirements). Audit completeness should have no effect on the quarterly EA GAAP disclosures as the auditor does not audit the interim financial statements. It is possible that audit quality could have an effect on the quarterly EA GAAP disclosures because of auditor involvement throughout the year (i.e., audit planning, interim testing, Section 404(b) internal control testing, etc.), but the effect should be smaller. I estimate separate models for the Q4/annual and quarterly samples (available sample of 57,065 observations) and test differences in coefficients across the two models using seemingly unrelated estimation. In untabulated results, I find that the coefficient on *AUDCOMP* is only significant in the Q4/annual model ($p < 0.01$) and the difference in coefficients is significant (χ^2 $p < 0.01$). I find positive and significant coefficients on *BIGN* and *ABFEES* in both the Q4/annual and Q1 to Q3 models ($p < 0.01$ in both models). However, the effect is significantly larger in Q4 for *ABFEES* (χ^2 $p < 0.05$). This falsification test provides evidence that the audit completeness effect is solely in Q4 and the audit quality effect is stronger in Q4.

Implications of Sarbanes-Oxley Act

Many key SOX provisions and other audit regulatory changes have the potential to change the role of the external auditor in management's total EA GAAP disclosures. To evaluate this, I reestimate Model (1) by including an indicator variable (*POST*) that uses the November 15, 2004 effective date for AS2/3 as the inflection point with post- (pre-) regulation observations coded 1 (0) and interacting this variable with the audit completeness and quality measures. In untabulated results, the coefficients on the interaction terms for audit completeness (*AUDCOMP* $*$ *POST*) and the audit size measure (*BIGN* $*$ *POST*) are not significant, suggesting no incremental change from the pre- to the post-period. However, the interaction term on abnormal audit fees (*ABFEES* $*$ *POST*) is positive and significant (Coeff. $= 0.1841$; t-stat $= 2.07$; $p < 0.05$), consistent with the relationship between *ABFEES* and total GAAP disclosures being incrementally stronger post-AS2/3.

V. MARKET IMPLICATIONS

In this section, I examine the market implications of more detailed EA GAAP disclosures by examining cross-sectional differences in EA information content (Beaver 1968; Landsman and Maydew 2002). I first confirm prior research that documents that EAs are more informative when they: (1) have more detailed GAAP disclosures (Francis et al. 2002; Collins et

[24] The primary audit completeness results may be a result of absolute completion (i.e., complete versus incomplete at EA release date) or the relative days leading up to the audit report date (i.e., EA released 25 days versus one day before the audit report date). I reestimate Model (1) replacing *AUDCOMP* with the dichotomous version of the measure (*AUDEA*). I next reestimate Model (1) (with *AUDCOMP*) for the subsample of companies that release the EA before the audit report date. Results for both sensitivity analyses are consistent with the results reported in Tables 5 and 6.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

TABLE 6

Multivariate Results for Individual Financial Statement GAAP Disclosures

Panel A: Results for Full Sample

	Pred. Sign	Income Statement Ratio (DR_{IS}) (1) Coeff. (t-stat)	Balance Sheet Ratio (DR_{BS}) (2) Coeff. (t-stat)	Cash Flow Stmt. Ratio (DR_{CF}) (3) Coeff. (t-stat)
AUDCOMP	+	0.0001	0.0010***	0.0013***
		(1.10)	(4.60)	(4.94)
BIGN	+	0.0160***	0.0425***	0.0523***
		(4.64)	(3.68)	(4.05)
ABFEES	+	0.0509*	0.5163***	0.4107***
		(1.88)	(5.44)	(3.66)
Controls, Year FE, Industry FE		Yes	Yes	Yes
Observations		22,278	22,278	22,278
Adjusted R^2		0.139	0.207	0.250

Panel B: Results for the Audit Completeness PSM Sample

	Pred. Sign	Income Statement Ratio (DR_{IS}) (1) Coeff. (t-stat)	Balance Sheet Ratio (DR_{BS}) (2) Coeff. (t-stat)	Cash Flow Stmt. Ratio (DR_{CF}) (3) Coeff. (t-stat)
AUDCOMP	+	−0.0000	0.0007***	0.0007***
		(−0.57)	(2.85)	(2.80)
BIGN	+	0.0190***	0.0534***	0.0452***
		(3.77)	(3.33)	(3.35)
ABFEES	+	0.1229***	0.5785***	0.4909***
		(3.32)	(5.00)	(4.67)
Controls, Year FE, Industry FE		Yes	Yes	Yes
Observations		7,508	7,508	7,508
Adjusted R^2		0.113	0.185	0.190

Panel C: Results for the Audit Quality PSM Sample

	Pred. Sign	Income Statement Ratio (DR_{IS}) (1) Coeff. (t-stat)	Balance Sheet Ratio (DR_{BS}) (2) Coeff. (t-stat)	Cash Flow Stmt. Ratio (DR_{CF}) (3) Coeff. (t-stat)
AUDCOMP	+	0.0003**	0.0011***	0.0015***
		(2.31)	(2.60)	(2.89)
BIGN	+	0.0078**	0.0380***	0.0197
		(2.06)	(3.06)	(1.34)
ABFEES	+	0.1188**	0.5792***	0.3339*
		(2.46)	(3.88)	(1.96)
Controls, Year FE, Industry FE		Yes	Yes	Yes
Observations		5,762	5,762	5,762
Adjusted R^2		0.199	0.242	0.199

(continued on next page)

TABLE 6 (continued)

Panel D: Results for Change Model (i.e., First-Differencing)

	Pred. Sign	Income Statement Ratio (ΔDR_{IS}) (1) Coeff. (t-stat)	Balance Sheet Ratio (ΔDR_{BS}) (2) Coeff. (t-stat)	Cash Flow Stmt. Ratio (ΔDR_{CF}) (3) Coeff. (t-stat)
$\Delta AUDCOMP$	+	−0.0000	0.0001*	0.0002***
		(−0.20)	(1.92)	(3.13)
$\Delta BIGN$	+	0.0018	0.0123**	0.0049
		(0.51)	(2.30)	(1.42)
$\Delta ABFEES$	+	0.0016	0.0500	0.0440**
		(0.07)	(1.26)	(2.04)
Controls, Year FE, Industry FE			Yes	Yes
Observations		19,115	19,115	19,115
Adjusted R^2		0.005	0.011	0.010

*, **, *** Indicate $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively, two-tailed tests.
Standard errors are clustered by company to compute t-statistics. The sample period includes years 2001 to 2011. Columns (1), (2), and (3) present the income statement, balance sheet, and cash flow model results, respectively. Panel A presents the results of the full sample. Panel B presents the results of the audit completeness (i.e., EA released on or after the audit report date versus prior to audit report date) propensity score match sample. Panel C presents the results of the audit quality (i.e., Big N versus non-Big N) propensity score match sample. Panel D presents the results of the change model (i.e., first-differencing), where the dependent, independent, and control variables are transformed to be the difference between the values in years *t* and *t*−1. Control variables are included, but not presented for brevity.
Variable definitions can be found in Appendix A.

al. 2009); (2) have complete audits (Marshall et al. 2015); and (3) have higher quality audits (Teoh and Wong 1993). I then extend this literature by examining how complete audits and higher quality audits moderate the information content increase provided by more detailed GAAP disclosures.

To the extent that the information content of the EA is increasing in the amount of GAAP disclosure (e.g., Francis et al. 2002; Collins et al. 2009; Pawlewicz 2011), audit completeness and quality have implications for the market through more detailed GAAP disclosure. The status of the audit (i.e., complete versus incomplete audit)[25] and the identity of the audit firm (Big N versus non-Big N audit firm) are known by the market as of the EA release date. Thus, complete audits and higher quality audits should not only have a main effect on EA information content, but also a moderating effect on the information content provided by more detailed GAAP disclosure. Consistent with prior research, I expect both a completed audit and a higher quality audit to have a positive main effect on the information content of the EA (Teoh and Wong 1993; Marshall et al. 2015).[26]

Completed audits may have a positive moderating effect on the information content of EA GAAP disclosures to the extent that the market perceives the release and/or GAAP information included therein to be more credible. However, EA releases with completed audits tend to be less timely (i.e., released later after year-end), potentially resulting in a negative moderating effect to the extent that investors place greater reliance on earlier alternative information sources to infer the GAAP information. I expect higher quality audits to have a positive moderating effect on EA GAAP disclosures.

EA Information Content Models

I estimate the moderating effect of audit completion/quality on the impact of GAAP disclosures on EA information content by adapting models from Landsman and Maydew (2002) and Collins et al. (2009). The sample is reduced to 19,501 observations due to missing data and the requirement that the 10-K filing not occur during the three-day window surrounding the EA date. Variables are defined in Appendix A and discussed further below:

[25] Marshall et al. (2015) verify that companies mark annual GAAP information as "Unaudited" when the annual EA is released prior to the audit report date (i.e., incomplete) and that no such tagline is included when the annual EA is released on or after the audit report date (i.e., complete).

[26] Teoh and Wong (1993) find that earnings response coefficients are higher for companies audited by Big N audit firms. Marshall et al. (2015) find that completed audits at the earnings release date are associated with lower (higher) bid/ask spreads (earnings response coefficients). I extend their study by providing evidence that completed audits have a moderating effect on the information content of management's EA GAAP disclosures.

American Accounting Association

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

$$
\begin{aligned}
(1)\ ARET_{i,t} \quad &= \beta_0 + \beta_1 DR_{TOTi,t} + \beta_2 AUDEA/BIGN_{i,t} + \beta_3 DR_{TOT} * AUDEA/BIGN_{i,t} + \beta_4 LNMVE_{i,t} \\
(2)\ AVOL_{i,t} \quad &\quad + \beta_5 EA_LAG_{i,t} + \beta_6 MBR_{i,t} + \beta_7 INTEXP_{i,t} + \beta_8 SPECITM_{i,t} + \beta_9 LOSS_{i,t} \\
(3)\ AVAR_{i,t} \quad &\quad + \beta_{10} UE_{i,t} + \beta_{11} LIT_{i,t} + year\ fixed\ effects + industry\ fixed\ effects + \varepsilon_{i,t} \qquad (5)
\end{aligned}
$$

I measure EA information content in three ways: (1) abnormal stock return volatility (*ARET*); (2) abnormal trading volume (*AVOL*); and (3) abnormal stock price volatility (*AVAR*) (Beaver 1968; Stice 1991; Landsman and Maydew 2002; Collins et al. 2009). The three measures examined together provide a better understanding of whether audit completion and higher quality audits moderate the effect of more detailed GAAP disclosures on the market's expectation of firm value (price reaction) and differential belief revision (volume reaction).

The three variables of interest are the total GAAP disclosure ratio (DR_{TOT}), an indicator variable for whether the audit is complete at the EA date (*AUDEA*), and auditor size (*BIGN*). I replace *AUDCOMP* with *AUDEA,* as investors only know whether an audit has already been completed and are not aware of how long it will take for an auditor to finish an audit that is incomplete as of the EA date. I also exclude the measure of abnormal fees, as current-period audit fees are not known to the market until they are subsequently disclosed in the proxy statement after the 10-K filing. I first estimate the model with DR_{TOT} to confirm prior research that has shown information content is increasing in GAAP disclosures. I then include the *AUDEA* (*BIGN*) variable to estimate the main effect of complete audits (audit quality) on information content. I finally include the interaction term $DR_{TOT} * AUDEA$ ($DR_{TOT} * BIGN$) to estimate the moderating effect of complete audits (audit quality). I include controls for company size (*LNMVE*), timing of the earnings release (*EA_LAG*), conservatism (*MBR*), interest expense (*INTEXP*), loss years (*LOSS*), existence of nonrecurring items (*SPECITM*), litigation risk (*LIT*), unexpected earnings (*UE*), year effects, and industry effects (Landsman and Maydew 2002).

Results of Information Content Analyses

Table 7 presents the results examining the total GAAP disclosures and EA information content. Columns (1), (2), and (3) present the results of abnormal returns, volume, and stock price volatility, respectively. I find positive and significant coefficients on DR_{TOT} across all three models ($p < 0.05$ for *ARET* and $p < 0.01$ for *AVOL* and *AVAR*). This is consistent with prior research that EA information content is increasing in GAAP disclosures (Francis et al. 2002). Given the earlier results of a positive association between audit completeness/quality and EA GAAP disclosure, these results are consistent with audit completeness and quality having market implications through management's EA GAAP disclosures.

Table 8, Panel A presents the main (Columns (1), (3), and (5)) and moderating effects (Columns (2), (4), and (6)) of complete audits. With respect to the main effect, the coefficient on *AUDEA* is positive and significant ($p < 0.05$) in the abnormal stock return model (*ARET*), consistent with complete audits being informative to the market's aggregate expectation of firm value. However, there is no effect on abnormal volume or stock price volatility as the coefficients on *AUDEA* are not significant. The coefficients on DR_{TOT} in Columns (2), (4), and (6) and the sum of the coefficients on DR_{TOT} and $DR_{TOT} * AUDEA$ are positive and significant across all three models, confirming that information content is increasing in GAAP disclosures for both complete and incomplete audits. Turning to the moderating effect, the coefficients on $DR_{TOT} * AUDEA$ are negative in all three models and significant for the abnormal volume and stock price volatility models ($p < 0.05$ for *AVOL;* $p < 0.10$ for *AVAR*). This suggests that complete audits attenuate the information content provided by additional GAAP disclosures.

To examine whether the timeliness of EA disclosures explains the attenuating effect, I partition the full sample into two separate subsamples based on whether the audit was complete or incomplete at the EA release date. I then perform a one-to-one match of observations from both subsamples based on *EA_LAG,* two-digit SIC code, and fiscal year to ensure that the treatment and control groups have statistically similar EA release dates. This results in a sample of 8,306 observations (4,153 matched pairs). In untabulated results, the coefficients on $DR_{TOT} * AUDEA$ are no longer statistically significant. This suggests that the attenuating effect of complete audits on the information content of more detailed GAAP disclosure is due to differential timing of the EA release.

Table 8, Panel B presents the main (Columns (1), (3), and (5)) and moderating effects (Columns (2), (4), and (6)) of audit quality. With respect to the main effect, the coefficient on *BIGN* is positive and significant ($p < 0.01$) in the abnormal stock price volatility model (*AVAR*), consistent with audit quality being informative to the market's aggregate expectation of firm value. However, the coefficients on *BIGN* in the abnormal stock return and volume models are not significant. The coefficients on the sum of DR_{TOT} and $DR_{TOT} * BIGN$ in Columns (2), (4), and (6) are positive and significant; however, the coefficients on DR_{TOT} are not significant. This suggests that the information content of GAAP disclosures is only increasing for Big N audits. Turning to the moderating effect, the coefficients are positive in all three models and significant in the abnormal stock return and volume models ($p < 0.05$ for *AVAR* and *AVOL*), consistent with audit quality enhancing the information content of expanded GAAP disclosures.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

TABLE 7

Multivariate Results Examining Total GAAP Disclosures and EA Information Content

Variables	Pred. Sign	(1) Abnormal Stock Returns (DV = *ARET*) Coeff. (t-stat)	(2) Abnormal Volume (DV = *AVOL*) Coeff. (t-stat)	(3) Abnormal Stock Price Volatility (DV = *AVAR*) Coeff. (t-stat)
DR_{TOT}	+	0.0060**	0.5574***	2.3184***
		(2.02)	(5.89)	(5.11)
LNMVE	−/+	−0.0012***	0.3370***	0.3964***
		(−2.59)	(23.32)	(6.10)
EA_LAG	−	−0.0002***	−0.0083***	−0.0347***
		(−4.00)	(−5.58)	(−5.02)
MBR	−/+	−0.0002*	0.0043	−0.0019
		(−1.93)	(1.54)	(−0.15)
INTEXP	−	0.0169	−0.4939	−2.6477
		(0.47)	(−0.66)	(−0.83)
SPECITM	±	−0.0005	−0.0740*	0.0629
		(−0.37)	(−1.88)	(0.33)
LOSS	−	−0.013***	−0.4419***	−1.9294***
		(−7.96)	(−10.12)	(−9.34)
UE	+	0.0065***	0.0617	−0.5133***
		(2.64)	(1.61)	(−2.87)
LIT	±	−0.0004	0.1832***	0.0314
		(−0.17)	(2.57)	(0.09)
Intercept		0.0105	−0.4561	4.6065
		(0.49)	(−0.75)	(1.43)
Year FE		Yes	Yes	Yes
Industry FE		Yes	Yes	Yes
Observations		19,501	19,501	19,501
Adj. R^2		0.014	0.164	0.065

*, **, *** Indicate $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively, two-tailed t-statistic tests.
Standard errors are clustered by company to compute t-statistics. The sample period is 2001 to 2011.
Variable definitions can be found in Appendix A.

VI. CONCLUSION

This study examines the role of the external audit in management's EA GAAP disclosure decisions. The EA is a key financial disclosure that has been shown to be more informative to investors than the 10-K filing. Yet, there are no regulations requiring the historical GAAP information included in the EA to be certified by the auditor. Recent trends indicate that the majority of EA disclosures are now issued before the completion of the year-end audit. Prior research has focused primarily on the role of auditing in improving the quality of financial statements included in the 10-K filing; however, there is relatively little research examining how the external audit impacts management's voluntary disclosures, especially in an EA setting. More complete and higher quality audits should help facilitate detailed GAAP disclosures, as the external audit increases management's confidence in the financial reporting system and the underlying financial statement balances. Furthermore, more EA GAAP disclosures, complete audits, and higher quality audits should have market implications by increasing the information content of the EA.

The results yield several key findings. I find that more complete audits at the earnings release date and higher quality audits are associated with management providing more detailed overall GAAP disclosures, driven primarily by management providing more detailed balance sheet and cash flow statement disclosures. These results are generally robust to concerns about the endogenous choice of EA timing in relation to audit completion, audit firm identity, and GAAP disclosure details. However, I acknowledge that these concerns can never be ruled out completely. I also find that complete audits and higher quality audits attenuate (enhance) the information content of more detailed GAAP disclosures.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

TABLE 8

Multivariate Results Examining Main and Moderating Effect of Audit Completeness/Quality on EA Information Content

Panel A: Main and Moderating Effect of Audit Completeness

		Abnormal Stock Returns (DV = *ARET*)		Abnormal Volume (DV = *AVOL*)		Abnormal Stock Price Volatility (DV = *AVAR*)	
	Pred. Sign	(1) Coeff. (t-stat)	(2) Coeff. (t-stat)	(3) Coeff. (t-stat)	(4) Coeff. (t-stat)	(5) Coeff. (t-stat)	(6) Coeff. (t-stat)
DR_{TOT}	+	0.0059**	0.0077**	0.5579***	0.6454***	2.3173***	2.6794***
		(2.00)	(2.37)	(5.89)	(5.89)	(5.10)	(5.03)
AUDEA	+	0.0039**	0.0075**	−0.0283	0.1491	0.0630	0.7965*
		(2.17)	(2.06)	(−0.61)	(1.51)	(0.30)	(1.74)
$DR_{TOT} * AUDEA$	±		−0.0071		−0.3434**		−1.4198*
			(−1.13)		(−1.98)		(−1.68)
Controls		Yes	Yes	Yes	Yes	Yes	Yes
Year FE		Yes	Yes	Yes	Yes	Yes	Yes
Industry FE		Yes	Yes	Yes	Yes	Yes	Yes
Observations		19,501	19,501	19,501	19,501	19,501	19,501
Adjusted R^2		0.015	0.015	0.164	0.165	0.065	0.065
		Sum of Coeff. (F-Stat)		Sum of Coeff. (F-Stat)		Sum of Coeff. (F-Stat)	
$DR_{TOT} + DR_{TOT} * AUDEA$		0.0006#		0.3020###		1.2596###	
		(2.81)		(18.11)		(13.40)	

Panel B: Main and Moderating effect of Audit Quality

		Abnormal Stock Returns (DV = *ARET*)		Abnormal Volume (DV = *AVOL*)		Abnormal Stock Price Volatility (DV = *AVAR*)	
	Pred. Sign	(1) Coeff. (t-stat)	(2) Coeff. (t-stat)	(3) Coeff. (t-stat)	(4) Coeff. (t-stat)	(5) Coeff. (t-stat)	(6) Coeff. (t-stat)
DR_{TOT}	±	0.0059**	−0.0068	0.5547***	0.0995	2.2574***	0.8453
		(2.00)	(−0.93)	(5.86)	(0.48)	(4.97)	(0.85)
BIGN	+	0.0007	−0.0075	0.0371	−0.2531*	0.8338***	−0.0666
		(0.32)	(−1.63)	(0.63)	(−1.94)	(3.13)	(−0.11)
$DR_{TOT} * BIGN$	+		0.0154		0.5504**		1.7078
			(1.99)**		(2.43)		(1.57)
Controls		Yes	Yes	Yes	Yes	Yes	Yes
Year FE		Yes	Yes	Yes	Yes	Yes	Yes
Industry FE		Yes	Yes	Yes	Yes	Yes	Yes
Observations		19,501	19,501	19,501	19,501	19,501	19,501
Adjusted R^2		0.014	0.015	0.164	0.165	0.065	0.065
		Sum of Coeff. (F-Stat)		Sum of Coeff. (F-Stat)		Sum of Coeff. (F-Stat)	
$DR_{TOT} + DR_{TOT} * BIGN$		0.0086##		0.6499###		2.5531###	
		(4.48)		(19.55)		(13.29)	

*, **, *** Indicate $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively, two-tailed t-statistic tests.
#, ##, ### Indicate $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively, F-statistic tests.
Standard errors are clustered by company to compute t-statistics. The sample period is 2001 to 2011. Control variables are included, but not presented for brevity. The sum of the coefficients on $DR_{TOT} + DR_{TOT} * AUDEA$ ($DR_{TOT} + DR_{TOT} * BIGN$) represents the effect of GAAP disclosures for the earnings releases when the audit is complete (audited by a Big N audit firm).
Variable definitions can be found in Appendix A.

These findings have important regulatory implications. The focus of the PCAOB, through its inspection/enforcement and standard-setting authority, is to improve the quality of public company audits and to ensure higher quality financial reporting for stakeholders. However, PCAOB standards (i.e., AS2/3) have resulted in less timely audits, resulting in an overwhelming majority of companies releasing EAs before the audit is substantially complete. It is important that regulators, especially the PCAOB, continue to evaluate the role of the external audit in all key financial disclosures and evaluate the impact that their standards have on the timeliness of audits and the capital markets.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

REFERENCES

Baber, W. R., S. Chen, and S. Kang. 2006. Stock price reaction to evidence of earnings management: Implications for supplementary financial disclosure. *Review of Accounting Studies* 11: 5–19.

Bagnoli, M., W. Kross, and S. G. Watts. 2002. The information in management's expected earnings report date: A day late, a penny short. *Journal of Accounting Research* 40 (5): 1275–1296.

Ball, R., S. Jayaraman, and L. Shivakumar. 2012. Audited financial reporting and voluntary disclosure as complements: A test of the confirmation hypothesis. *Journal of Accounting and Economics* 53: 136–166.

Beaver, W. 1968. The information content of annual earnings announcements. *Journal of Accounting Research* 6: 67–91.

Becker, C. L., M. L. DeFond, J. Jiambalvo, and K. R. Subramanyam. 1998. The effect of audit quality on earnings management. *Contemporary Accounting Research* 15 (1): 4–24.

Begley, J., and P. E. Fischer. 1998. Is there information in an earnings announcement delay? *Review of Accounting Studies* 3: 347–363.

Belsley, D. A., E. Kuh, and R. E. Welsch. 1980. *Regression Diagnostics: Identifying Influential Data and Sources of Collinearity*. New York, NY: John Wiley & Sons.

Beyer, A., D. A. Cohen, T. Z. Lys, and B. R. Walther. 2010. The financial reporting environment: Review of the recent literature. *Journal of Accounting and Economics* 50: 296–343.

Blankley, A. I., D. N. Hurtt, and J. E. MacGregor. 2012. Abnormal audit fees and restatements. *Auditing: A Journal of Practice & Theory* 31 (1): 79–96.

Bochner, S. E., and R. C. Blake. 2008. The earnings release: Legal requirements and best practices. *Insights: The Corporate Securities Law Advisor* 22 (3): 1–16.

Botosan, C. 1997. Disclosure level and the cost of equity capital. *The Accounting Review* 72: 323–349.

Botosan, C., and M. Harris. 2000. Motivations for a change in disclosure frequency and its consequences: An examination of voluntary quarterly segment disclosures. *Journal of Accounting Research* 38: 329–353.

Botosan, C., and M. Plumlee. 2002. A re-examination of disclosure level and the expected cost of equity capital. *Journal of Accounting Research* 40: 21–40.

Bronson, S. N., C. E. Hogan, M. F. Johnson, and K. Ramesh. 2011. The unintended consequences of PCAOB Auditing Standard Nos. 2 and 3 on the reliability of preliminary earnings releases. *Journal of Accounting and Economics* 51: 95–114.

Bronson, S. N., A. Masli, and J. H. Schroeder. 2015. *Jumping the Gun: Consequences of Announcing Annual Earnings When the Audit is Less Complete*. Working paper, Indiana University.

Bushee, B. J., D. A. Matsumoto, and G. S. Miller. 2002. Open versus closed conference calls: The determinants and effects of broadening access to disclosure. *Journal of Accounting and Economics* 34: 149–180.

Cattell, R. B., and S. Vogelman. 1977. A comprehensive trial of the scree and KG criteria for determining the number of factors. *Multivariate Behavioral Research* 12: 289–325.

Causholli, M., M. DeMartinis, D. Hay, and W. Knechel. 2010. Audit markets, fees and production: Towards an integrated view of empirical audit research. *Journal of Accounting Literature* 30: 167–215.

Chartered Financial Analyst Institute Centre for Financial Market Integrity/Business Roundtable Institute for Corporate Ethics. 2007. *Apples to Apples: A Template for Reporting Quarterly Earnings*. Available at: http://www.corporate-ethics.org/pdf/earnings_release_template.pdf

Chen, S., M. L. DeFond, and C. W. Park. 2002. Voluntary disclosure of balance sheet information in quarterly earnings announcements. *Journal of Accounting and Economics* 33: 229–251.

Choi, J., J. Kim, and Y. Zang. 2010. Do abnormally high audit fees impair audit quality? *Auditing: A Journal of Practice & Theory* 29 (2): 115–140.

Collins, D., O. Li, and H. Xie. 2009. What drives the increased informativeness of earnings announcements over time? *Review of Accounting Studies* 14: 1–30.

Cutler, J., A. Davis, and K. Peterson. 2013. *Voluntary Disclosure and the Outcome of Securities Litigation*. Working paper, University of Oregon.

DeAngelo, L. 1981. Auditor size and audit quality. *Journal of Accounting and Economics* 3: 183–199.

DeFond, M., and J. Zhang. 2014. A review of archival auditing research. *Journal of Accounting and Economics* 58: 275–326.

Deloitte. 2012. *Earnings Guidance*. Available at: http://www.corpgov.deloitte.com/site/caneng/financial-reporting/transparency/earnings-guidance/

American Accounting Association

Desai, H., C. E. Hogan, and M. S. Wilkins. 2006. The reputational penalty for aggressive accounting: Earnings restatements and management turnover. *The Accounting Review* 81 (1): 83–112.

Diamond, C. J., and I. Yevmenenko. 2011. Earnings releases and earnings calls. *Practical Law Journal* (October): 35–44.

Diamond, D. W. 1985. Optimal release of information by firms. *Journal of Finance* 40: 1071–1094.

Dopuch, N., and D. Simunic. 1980. The nature of competition in the auditing profession: A descriptive and normative view. In *Regulation and the Accounting Profession,* edited by Buckley, J., and F. Weston, 34 (2): 283–289. Belmont, CA: Lifetime Learning Publications.

Doyle, J., and M. Magilke. 2013. Decision usefulness and accelerated filing deadlines. *Journal of Accounting Research* 51: 549–581.

D'Souza, J., K. Ramesh, and M. Shen. 2010. Disclosure of GAAP line items in earnings announcements. *Review of Accounting Studies* 15: 179–219.

Dunn, K. A., and B. W. Mayhew. 2004. Audit firm industry specialization and client disclosure quality. *Review of Accounting Studies* 9: 35–58.

Einhorn, E., and A. Ziv. 2008. Intertemporal dynamics of corporate voluntary disclosures. *Journal of Accounting Research* 46 (3): 567–589.

Evans, M. E. 2015. Commitment and cost of equity capital: An examination of timely balance sheet disclosure in earnings announcements. *Contemporary Accounting Research* (forthcoming).

Field, L., M. Lowry, and S. Shu. 2005. Does disclosure deter or trigger litigation? *Journal of Accounting and Economics* 39: 487–507.

Francis, J. R. 2004. What do we know about audit quality? *British Accounting Review* 36: 345–368.

Francis, J. R. 2011. A framework for understanding and researching audit quality. *Auditing: A Journal of Practice & Theory* 30 (2): 125–152.

Francis, J. R., and M. Yu. 2009. The effect of Big 4 office size on audit quality. *The Accounting Review* 84 (5): 1521–1552.

Francis, J. R., K. Schipper, and L. Vincent. 2002. Expanded disclosures and the increased usefulness of earnings announcements. *The Accounting Review* 77: 515–546.

Frankel, R., M. Johnson, and D. Skinner. 1999. An empirical examination of conference calls as a voluntary disclosure medium. *Journal of Accounting Research* 37: 133–150.

Givoly, D., and D. Palmon. 1982. Timeliness of annual earnings announcements: Some empirical evidence. *The Accounting Review* 57: 486–508.

Graham, J. R., C. R. Harvey, and S. Rajgopal. 2005. The economic implications of corporate financial reporting. *Journal of Accounting and Economics* 40: 3–73.

Guo, S., and M. Fraser. 2010. *Propensity Score Analysis: Statistical Methods and Applications*. Thousand Oaks, CA: Sage Publications, Inc.

Hay, D. C., W. R. Knechel, and N. Wong. 2006. Audit fees: A meta-analysis of the effects of supply and demand. *Contemporary Accounting Research* 23 (1): 141–191.

Healy, P. M., and K. G. Palepu. 2001. Information asymmetry, corporate disclosure, and the capital markets: A review of the empirical disclosure literature. *Journal of Accounting and Economics* 31: 405–440.

Hogan, C. E., and M. S. Wilkins. 2008. Evidence on the audit risk model: Do auditors increase audit fees in the presence of internal control deficiencies? *Contemporary Accounting Research* 25 (1): 219–242.

Kinney, W. R., and R. Libby. 2002. Discussion of the relation between auditors' fees for nonaudit services and earnings management. *The Accounting Review* 77 (Supplement): 107–114.

Kothari, S. P. 2001. Capital markets research in accounting. *Journal of Accounting and Economics* 31: 105–231.

Krishnan, J., and J. S. Yang. 2009. Recent trends in audit reports and earnings announcement lags. *Accounting Horizons* 23 (3): 265–288.

Kross, W. 1981. Earnings and announcement time lags. *Journal of Business Research* 9: 267–281.

Kross, W., and D. A. Schroeder. 1984. An empirical investigation of the effect of quarterly earnings announcement timing on stock returns. *Journal of Accounting Research* 22 (1): 153–176.

Lambert, T. A., K. L. Jones, J. F. Brazel, and S. Showalter. 2015. *Audit Time Pressure and Earnings Quality: An Examination of Accelerated Filings*. Working paper, George Mason University, Lehigh University, and North Carolina State University.

Landsman, W. R., and E. L. Maydew. 2002. Has the information content of quarterly earnings announcements declined in the past three decades? *Journal of Accounting Research* 40: 797–808.

Lang, M., and R. Lundholm. 1993. Cross-sectional determinants of analyst ratings of corporate disclosures. *Journal of Accounting Research* 31: 246–271.

Lawrence, A., M. Minutti-Meza, and P. Zhang. 2011. Can Big 4 versus non-Big 4 differences in audit-quality proxies be attributed to client characteristics? *The Accounting Review* 86 (1): 259–286.

Li, E. X., and K. Ramesh. 2009. Market reaction surrounding the filing of periodic SEC reports. *The Accounting Review* 84: 1171–1208.

Marshall, N., J. H. Schroeder, and T. L. Yohn. 2015. *Does an Incomplete versus a Completed Audit Affect the Market's Reliance on the Earnings Announcement?* Working paper, Indiana University.

Miao, B., S. H. Teoh, and Z. Zhu. 2013. *The Effect of Salience on Market Valuation of Accruals: The Effect of Statement of Cash Flow Disclosure in Earnings Announcements*. Working paper, University of California, Irvine.

National Investor Relations Institute (NIRI). 2012. *Standards of Practice on Earnings Release Content*. Alexandria, VA: NIRI.

Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.
The Accounting Review 2016.91:677-705.

Pawlewicz, R. 2011. *The Economic Consequences of Regulation G: Changes in Earnings Announcements Timing and Investor Responses*. Working paper, George Mason University.

Public Company Accounting Oversight Board (PCAOB). 2004a. *An Audit of Internal Control over Financial Reporting Performed in Conjunction with an Audit of Financial Statements*. Auditing Standard No. 2. Washington, DC: PCAOB. Available at: http://pcaobus.org/Standards/Auditing/Pages/Auditing_Standard_2.aspx

Public Company Accounting Oversight Board (PCAOB). 2004b. *Audit Documentation*. *Auditing*. Auditing Standard No. 3. Washington, DC: PCAOB. Available at: http://pcaobus.org/Standards/Auditing/Pages/Auditing_Standard_3.aspx

Public Company Accounting Oversight Board (PCAOB). 2007. *An Audit of Internal Control over Reporting that is Integrated with Audit of Financial Statements and Related Independence Rule and Conforming Amendments*. Auditing Standard No. 5. Washington, DC: PCAOB. Available at: http://pcaobus.org/Standards/Auditing/Pages/Auditing_Standard_5.aspx

Public Company Accounting Oversight Board (PCAOB). 2011. *What the PCAOB Expects for the Coming Year and Beyond*. Washington, DC: PCAOB: Available at: http://pcaobus.org/News/Speech/Pages/12052011_DotyAICPA.aspx

Rogers, J. L., and A. Van Buskirk. 2009. Shareholder litigation and changes in disclosure behavior. *Journal of Accounting and Economics* 47: 136–156.

Rosenbaum, P. R., and D. B. Rubin. 1983. The central role of the propensity score in observational studies for causal effects. *Biometrika* 70 (April): 41–55.

Scott, T. W. 1994. Incentives and disincentives for financial disclosure: Voluntary disclosure of defined benefit pension plan information by Canadian firms. *The Accounting Review* 69: 26–43.

Securities and Exchange Commission (SEC). 2002. *Proposed Rule: Acceleration of Periodic Report Filing Dates and Disclosure Concerning Website Access to Reports*. Release No. 33-8089. Available at: http://www.sec.gov/rules/proposed/33-8089.htm

Securities and Exchange Commission (SEC). 2003. *Final Rule: Acceleration of Periodic Report Filing Dates and Disclosure Concerning Website Access to Reports*. Release No. 33-8128. Available at: http://www.sec.gov/rules/final/33-8128.htm

Securities and Exchange Commission (SEC). 2008. *Final Report of the Advisory Committee on Improvements to Financial Reporting (CIFiR)*. Available at: http://www.sec.gov/about/offices/oca/acifr/acifr-finalreport.pdf

Sengupta, P. 2004. Disclosure timing: Determinants of quarterly earnings release dates. *Journal of Accounting and Public Policy* 23: 457–482.

Simunic, D. 1980. The pricing of audit services: Theory and evidence. *Journal of Accounting Research* 18 (Spring): 161–190.

Steinberg, M. I. 2009. *Securities Regulation*. Revised fifth edition. New Providence, NJ: LexisNexis Group.

Stempel, J. 2012. *Groupon Sued by Investor over Revised Results, IPO*. Reuters (April 3). Available at: http://www.reuters.com/article/2012/04/04/us-groupon-lawsuit-idUSBRE83303Y20120404

Stice, E. 1991. The market reaction to 10-K and 10-Q filings and to subsequent *The Wall Street Journal* earnings announcements. *The Accounting Review* 66: 42–55.

Tang, M. 2012. *What Guides the Guidance? An Empirical Examination of Dynamic Disclosure Theory*. Working paper, University of Rochester.

Teoh, S., and T. Wong. 1993. Perceived auditor quality and the earnings response coefficient. *The Accounting Review* 68: 346–366.

Trueman, B. 1986. Why do managers voluntarily release earnings forecasts? *Journal of Accounting and Economics* 8: 53–71.

Verrecchia, R. E. 2001. Essays on disclosure. *Journal of Accounting and Economics* 32: 97–180.

Wasley, C. E., and J. S. Wu. 2006. Why do managers voluntarily issue cash flow forecasts? *Journal of Accounting Research* 44 (2): 389–429.

Wooldridge, J. M. 2002. *Econometric Analysis of Cross Section and Panel Data*. Cambridge, MA: MIT Press.

Yang, H. 2012. Capital market consequences of managers' voluntary disclosure styles. *Journal of Accounting and Economics* 53: 167–184.

Zhang v. Groupon. 2012. Case: 1:12–cv-02450. U.S. District Court, Northern District of Illinois, Eastern Division. Retrieved from Lexis-Nexis: April 11, 2012.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

American Accounting Association

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

APPENDIX A

Variable Definitions

Variable	Definition
Dependent Variables	
DR_{TOT}	Sum of total non-missing financial statement items per Compustat Preliminary History divided by sum of total non-missing items per Compustat Unrestated Quarterly. A list of the financial statement items used in the calculation is provided in Appendix B.
DR_{IS}	Sum of total non-missing income statement items per Compustat Preliminary History divided by sum of total non-missing items per Compustat Unrestated Quarterly. A list of the financial statement items used in the calculation is provided in Appendix B.
DR_{BS}	Sum of total non-missing balance sheet items per Compustat Preliminary History divided by sum of total non-missing items per Compustat Unrestated Quarterly. A list of the financial statement items used in the calculation is provided in Appendix B.
DR_{CF}	Sum of total non-missing cash flow statement items per Compustat Preliminary History divided by sum of total non-missing items per Compustat Unrestated Quarterly. A list of the financial statement items used in the calculation is provided in Appendix B.
ARET	Abnormal stock returns is calculated consistent with Collins et al. (2009). *ARET* is estimated as the three-day average market model residuals, $(RET3_{it} - \hat{a}_{1it} - \hat{a}_{2it}MKT3_t)1/3$, around the earnings announcement date (i.e. day 0) where $RET3_{it}$ and $MKT3_{it}$ are the three-day cumulative stock returns and value-weighted market returns for days −1 to +1, respectively. Coefficients $\hat{a}_1$ and $\hat{a}_2$ are estimated using a market model ($RET_{it} = a_1 + a_2MKT_{it} + \varepsilon_{it}$) for the two-year period surrounding the earnings announcement ($t = -232$ to -20 and $t = +20$ to $+232$).
AVOL	Abnormal trading volume is calculated consistent with Landsman and Maydew (2002). $AVOL_{it} = (\bar{A}_{it} - \bar{A}_i)/\sigma_{i,}$ where A_{it} is the total shares of firm i traded during day t, with t representing the three days (−1 to +1) surrounding the EA date (i.e. day 0), scaled by total shares outstanding for firm i during day t. $\bar{A}_i$ and σ_i represent the mean and standard deviation in trading volume for firm i for the two years surrounding the EA date ($t = -232$ to -20 and $t = +20$ to $+232$). The metric used is the simple sum of the *AVOL* over the announcement window (−1,+1).
AVAR	Abnormal stock price volatility is calculated consistent with Landsman and Maydew (2002). $AVAR_{it} = \delta^2{}_{it}/\sigma^2{}_i$ where $AVAR_{it}$ is calculated for day t, with t representing the three days (−1 to +1) surrounding the EA date (i.e. day 0) for firm i. $\delta_{it} = R_{it} - (a_{1i} + a_{2i}R_{mt})$, where R_{it} is the raw return of firm i for day t and R_{mt} is the equal weighted return of the market for day t. a_{1i} and a_{2i} are firm i's market model estimates and $\sigma^2{}_i$ is the variance of firm i's market model adjusted returns for the two-year period surrounding the EA date ($t = -232$ to -20 and $t = +20$ to $+232$). Estimation of market model parameters and Std. Dev. excludes 20 trading days on either side of the announcement. The metric used is the simple sum of *AVAR* over the announcement window (−1,+1).
Independent Variables	
AUDCOMP	The number of days between the earnings announcement date and the audit report date, truncated at 0 when the earnings announcement date occurs on or after the audit report date.
AUDEA	An indicator variable equal to 1 if the earnings announcement is issued on or after the audit report date, and 0 otherwise.
BIGN	An indicator variable equal to 1 if the company is audited by a Big N audit firm, and 0 otherwise.

(continued on next page)

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

APPENDIX A (continued)

Variable	Definition
ABFEES	The residual from the following regression based on Hogan and Wilkins (2008) scaled by the natural log of audit fees: $LOGFEE = \beta_1 LOGASSETS + \beta_2 LEVERAGE + \beta_3 ROA + \beta_4 GROWTH + \beta_5 LOSS + \beta_6 ARINV + \beta_7 MERGER + \beta_8 LNBSEG + \beta_9 FOREIGN + \beta_{10} GC + \beta_{11} BIGN + \beta_{12} INITIALAUD + \beta_{13} YE + \beta_{14} AUD_LAG + \beta_{15} OP_404b + \beta_{16} MW_302 +$ *industry fixed effects* + *year fixed effects* + ε. *LOGFEE* = natural log of total audit fees; *LOGASSETS* = natural log of total assets (AT); *LEVERAGE* = total liabilities (LT) divided by total assets (AT); *ROA* = net income before extraordinary items (IB) divided by average total assets (AT); *GROWTH* = current-year assets less prior-year assets divided by prior-year assets; *LOSS* = an indicator variable equal to 1 if income before extraordinary items is negative (IB), and 0 otherwise; *ARINV* = accounts receivable (RECT) plus inventory (INVT) divided by total assets (AT); *MERGER* = an indicator variable equal to 1 if the company is involved in a merger, and 0 otherwise; *LNBSEG* = natural log of the total number of business segments; *FOREIGN* = an indicator variable equal to 1 if the company reports foreign currency adjustment (FCA), and 0 otherwise; *GC* = an indicator variable equal to 1 if auditor issues a going concern opinion, and 0 otherwise; *BIGN* = an indicator variable equal to 1 if audited by a Big 4 audit firm, and 0 otherwise; *INITIALAUD* = an indicator variable equal to 1 if first year of auditor/client relationship, and 0 otherwise; *YE* = an indicator variable equal to 1 if client fiscal year ends December 31, and 0 otherwise; *AUD_LAG* = number of days between financial statement period end and the audit report date; *OP_404b* = an indicator variable equal to 1 if client receives a 404b opinion, and 0 otherwise; *MW_302* = an indicator variable equal to 1 if client reports a Section 302 internal control weakness, and 0 otherwise; *industry fixed effects* = indicator variables for each two-digit SIC code classification; and *year fixed effects* = indicator variables for the year of the observation.
Control Variables	
EARVZ	An indicator variable equal to 1 if the net income balance reported in the current-period earnings announcement is different than the net income balance reported in the 10-K filing, and 0 otherwise. Data are hand-verified by comparing Preliminary History and Unrestated Quarterly.
EA_LAG	Number of days between the fiscal year-end and the earnings announcement date (RDQ − DATADATE).
SUPDISC	Sum of six indicator variables: (1) HIGHTECH = 1 for companies with SIC codes 2833–2836 (Drugs), 8731–8734 (R&D services), 7371–7379 (programming), 3570–3577 (computers), 3600–3674 (electronics), or 3810–3845 (precise measurement instruments), and 0 otherwise; (2) LOSS = 1 if earnings before extraordinary items (IB) is negative for the year, and 0 otherwise; (3) ABSFE = 1 if the absolute value of expectation error (defined as reported EPS minus most recent consensus mean analysts' forecast for I/B/E/S companies or seasonally adjusted change in EPS if non-I/B/E/S companies) is greater than $0.01 during the current year, and 0 otherwise; (4) MERGER = 1 if the company engages in M&A activities during the current year, and 0 otherwise; (5) STDRET = 1 if the standard deviation of stock returns measured over the previous 250 days (at least 100 days of stock returns required) is above the median, and 0 otherwise; and (6) AGE = 1 if the company's age (from the first day reported in Compustat to the current period) is below the two-digit SIC industry median, and 0 otherwise.
LNASSETS	Natural log of total assets (AT).
ACCEL	An indicator variable equal to 1 if Audit Analytics identifies as an accelerated filer, and 0 otherwise. If data are missing in Audit Analytics, then I calculate market capitalization from Compustat (CSHO * PRCC_F), classifying those companies with market capitalization between $75 million to $700 million as 1, and 0 otherwise.
LGACCEL	An indicator variable equal to 1 if Audit Analytics identifies as a large accelerated filer, and 0 otherwise. If data are missing in Audit Analytics, then I calculate market capitalization from Compustat (CSHO * PRCC_F), classifying those companies with market capitalization greater than or equal to $700 million as 1, and 0 otherwise.
Industry FE	Indicator variables for the two-digit SIC codes.
Year FE	Indicator variables for each year in the sample.

(continued on next page)

American Accounting Association

APPENDIX A (continued)

Variable	Definition
Variables used to Construct Factor Scores (See Appendix C)	
LNMVE	Natural log of market value of equity (CSHO ∗ PRCC_F).
ANALYST_FOLL	Number of analysts following the company on the I/B/E/S database during the current year, and 0 if not on the I/B/E/S database.
ANALYST_COVG	An indicator variable equal to 1 if the company is followed by analysts on the I/B/E/S database during the current year, and 0 otherwise.
SHAREHOLDER	Natural log of the total number of common shareholders (CSHR).
INSTHOLD	Percentage of shares held by institutions, obtained from the Thomson Reuters 13-F database.
ROA	Income before extraordinary items (IB) divided by total assets (AT).
OCF	Total operating cash flows (OANCF) divided by total assets (AT).
UE_NEG	An indicator variable equal to 1 if income before extraordinary items (IB) for the current year is less than income before extraordinary items during the previous year (IB at time $t-1$), and 0 otherwise.
LOSS	An indicator variable equal to 1 if income before extraordinary items (IB) is less than 0 for the current year, and 0 otherwise.
SPECITM	An indicator variable equal to 1 if there is a special item disclosure in Compustat with a value (SPI), and 0 otherwise.
LIT	An indicator variable equal to 1 if the company is in a high litigation-risk industry, and 0 otherwise. High litigation-risk industries are defined as companies with SIC codes in the following industries: 2833–2836 (biotechnology); 3570–3577 and 7370–7374 (computers); 3600–3674 (electronics); and 5200–5961 (retailing).
VOLUME	Total number of shares traded over the year (from CRSP) divided by outstanding shares at the end of the year (CSHO).
STDRET	Standard deviation of stock returns (from CRSP) measured over the previous 250 days (a minimum of 100 days of stock returns is required).
LNBSEG	Natural log of the number of business segments reported in Compustat's Segment File.
MBR	Market value/net book value [(CSHO ∗ PRCC_F)/(AT − LT)].
MKTCONC	Herfindahl Index calculated for each two-digit SIC code using Compustat observations.
Variables in the Market Implications Model Not Defined Above	
INTEXP	Total interest expense (XINT) scaled by total assets (AT).
UE	Difference between income before extraordinary items (IB) in the current year and the prior year.
Variables in Propensity Score Match Models Not Defined Above	
LEV	Total liabilities (LT) divided by total assets (AT).
INVREC	Inventory (INVT) plus receivables (RECT) divided by end-of-year assets (AT).
MERGER	An indicator variable equal to 1 if the client discloses any merger or acquisition activity, and 0 otherwise. Obtained from the Compustat footnote file.
GC	An indicator variable equal to 1 if the audit opinion contains a going concern paragraph, and 0 otherwise. Obtained from Audit Analytics Opinion file.
YE	An indicator variable equal to 1 if the client has a calendar year end, and 0 otherwise (FYR).
ATURN	Sales (REVT) at time t divided by total assets (AT) at time $t-1$.
CURR	Current assets at time t (ACT) divided by current liabilities at time t (LCT).

Compustat data items are in parentheses.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

APPENDIX B

Disclosure Ratio Calculation

Variable Name	Preliminary History Quarterly	Unrestated Quarterly
Income Statement		
Selling, General, and Administrative Expenses	XSGAQ_P	XSGAQR
Sales (Net)	SALEQ_P	SALEQR
Minority Interest (Income Account)	MIIQ_P	MIIQR
R&D Expense	XRDQ_P	XRDQR
Depreciation and Amortization	DPQ_P	DPQR
Income Taxes—Total	TXTQ_P	TXTQR
Income Before Extraordinary Items	IBQ_P	IBQR
Income Before EI—Adjusted from CSE	IBADJQ_P	IBADJQR
Operating Income Before Depreciation	OIBDPQ_P	OIBDPQR
Interest Expense	XINTQ_P	XINTQR
Pretax Income	PIQ_P	PIQR
Dividends—Preferred	DVPQ_P	DVPQR
Income Before EI—Avail. for Common	IBCOMQ_P	IBCOMQR
EI and Discontinued Operations	XIDOQ_P	XIDOQR
Cost of Goods Sold	COGSQ_P	COGSQR
Non-Operating Income (Expense)	NOPIQ_P	NOPIQR
Special Items	SPIQ_P	SPIQR
Discontinued Operations	DOQ_P	DOQR
Deferred Taxes (Income Account)	TXDIQ_P	TXDIQR
Net Income (Loss)	NIQ_P	NIQR
Accounting Changes—Cum. Effect	ACCHGQ_P	ACCHGQR
Extraordinary Items	XIQ_P	XIQR
Common Stock Equity—Dollar Savings	CSTKEQ_P	CSTKEQR
Balance Sheet		
Cash and Short-Term Investments	CHEQ_P	CHEQR
Receivables—Total	RECTQ_P	RECTQR
Inventories—Total	INVTQ_P	INVTQR
Current Assets—Other	ACOQ_P	ACOQR
Current Assets—Total	ACTQ_P	ACTQR
Depreciation, Depletion and Amortization (Accum.)	DPACTQ_P	DPACTQR
Property, Plant, and Equip—Total (Net)	PPENTQ_P	PPENTQR
Assets—Other	AOQ_P	AOQR
Assets—Total	ATQ_P	ATQR
Debt in Current Liabilities	DLCQ_P	DLCQR
Accounts Payable	APQ_P	APQR
Income Taxes Payable	TXPQ_P	TXPQR
Current Liabilities—Other	LCOQ_P	LCOQR
Current Liabilities—Total	LCTQ_P	LCTQR
Liabilities—Other	LOQ_P	LOQR
Long-Term Debt—Total	DLTTQ_P	DLTTQR
Deferred Taxes and Inv. Tax Credits	TXDITCQ_P	TXDITCQR
Minority Interest	MIBQ_P	MIBQR
Liabilities—Total	LTQ_P	LTQR
Preferred Stock—Carrying Value	PSTKQ_P	PSTKQR
Common Stock	CSTKQ_P	CSTKQR
Capital Surplus	CAPSQ_P	CAPSQR
Retained Earnings	REQ_P	REQR
Common Equity—Total	CEQQ_P	CEQQR
Stockholders' Equity—Total	SEQQ_P	SEQQR
Preferred Stock—Redeemable	PSTKRQ_P	PSTKRQR
Treasury Stock—Total Dollar Amount	TSTKQ_P	TSTKQR
Property, Plant, and Equipment—Total (Gross)	PPEGTQ_P	PPEGTQR

(continued on next page)

American Accounting Association

APPENDIX B (continued)

Variable Name	Preliminary History Quarterly	Unrestated Quarterly
Statement of Cash Flows		
Cash and Cash Equity—Inc. (Dec.) (SCF)	CHECHQ_P	CHECHQR
Changes in Current Debt (SCF)	DLCCHQ_P	DLCCHQR
Income Before EI (SCF)	IBCQ_P	IBCQR
Depreciation and Amortization (SCF)	DPCQ_P	DPCQR
EI and Discontinued Operations (SCF)	XIDOCQ_P	XIDOCQR
Deferred Taxes (SCF)	TXDCQ_P	TXDCQR
Equity in Net Loss (Earnings) (SCF)	ESUBCQ_P	ESUBCQR
Funds from Operations—Other (SCF)	FOPOQ_P	FOPOQR
Sale of Property, Plant, and Equipment (SCF)	SSPEQ_P	SSPEQR
Sale of Common and Preferred Stock (SCF)	SSTKQ_P	SSTKQR
Sale of Investments (SCF)	SIVQ_P	SIVQR
Long-Term Debt—Issuance (SCF)	DLTISQ_P	DLTISQR
Cash Dividends (SCF)	DVQ_P	DVQR
Capital Expenditures (SCF)	CAPXQ_P	CAPXQR
Increase in Investments (SCF)	IVCHQ_P	IVCHQR
Long-Term Debt—Reduction (SCF)	DLTRQ_P	DLTRQR
Purchase of Com. and Preferred Stock (SCF)	PRSTKCQ_P	PRSTKCQR
Acquisitions (SCF)	AQCQ_P	AQCQR
Sale of PP&E and Sale of Investment (SCF)	SPPIVQ_P	SPPIVQR
Accounts Receivable—Dec. (Inc.) (SCF)	RECCHQ_P	RECCHQR
Inventory—Dec. (Inc.) (SCF)	INVCHQ_P	INVCHQR
Acct Payable and Accrued Liability (SCF)	APALCHQ_P	APALCHQR
Income Tax—Accrued—Inc. (Dec.) (SCF)	TXACHQ_P	TXACHQR
Assets and Liabilities—Other (NC) (SCF)	AOLOCHQ_P	AOLOCHQR
Operating Activities—Net CF (SCF)	OANCFQ_P	OANCFQR
Short-Term Investment—Change (SCF)	IVSTCHQ_P	IVSTCHQR
Investing Activities—Other (SCF)	IVACOQ_P	IVACOQR
Investing Activities—Net CF (SCF)	IVNCFQ_P	IVNCFQR
Financing Activities—Other (SCF)	FIAOQ_P	FIAOQR
Financing Activities—Net CF (SCF)	FINCFQ_P	FINCFQR
Exchange Rate Effect—(SCF)	EXREQ_P	EXREQR
Interest Paid—Net (SCF)	INTPNQ_P	INTPNQR
Income Taxes Paid (SCF)	TXPDQ_P	TXPDQR

The disclosure ratios from D'Souza et al. (2010) are based on a total of 84 data items that are collected by Compustat: 23 from the income statement, 28 from the balance sheet, and 33 from the cash flow statement. The numerator is the sum of the number of non-missing items in the Compustat Preliminary History Quarterly Database. The denominator is the sum of the number of non-missing items recorded in the Compustat Unrestated Quarterly Database from the initial 10-Q or 10-K filings. The table above lists the data items for each financial statement.

APPENDIX C

Factor Analysis for Control Variables

As discussed in Section III, I use principle component factor analysis to reduce the dimensionality of the 16 disclosure measures used in prior studies. The results are presented in Table 9. Consistent with D'Souza et al. (2010), I use promax oblique rotation to allow the extracted factors to be correlated, which is consistent with the relationships of the underlying constructs (i.e., bad news is likely correlated with litigation risk). I retain factors based on the Scree test, as well as the proportion of variance explained by the factors (Cattell and Vogelman 1977). I identify five factors that explain a combined 69.75 percent of the total variation in the 16 disclosure measures.

The first factor, which I label Information Environment (*INFOENVT*), loads positively on variables capturing market demand for information: market value of equity (*LNMVE*), number of analysts following the company (*ANALYST_FOLL*), whether the company is followed by analysts (*ANALYST_COVG*), and the proportion of shares held by institutional investors (*INSTHOLD*).The second factor, which I label Litigation Risk (*LITRISK*), loads positively on variables capturing whether the company is in a high litigation-risk industry (*LIT*), has high trading volume during the year (*VOLUME*), and has high stock

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

The Accounting Review 2016.91:677-705.
Downloaded from aaajournals.org by Kings College London-FWIC Journals on 09/20/19. For personal use only.

TABLE 9

Factor Analysis

Variables	Information Environment (*INFOENVT*)	Litigation Risk (*LITRISK*)	Bad News (*BADNEWS*)	Complexity (*COMPLX*)	*DFACT*
LNMVE	**0.5358**	0.0769	−0.1011	**0.5012**	−0.0495
ANALYST_FOLL	**0.7077**	0.2744	−0.0032	0.2017	0.0019
ANALYST_COVG	**0.8962**	−0.0786	0.1241	−0.1456	−0.0525
SHAREHOLDER	−0.0549	−0.0210	0.0475	**0.8097**	0.0433
INSTHOLD	**0.8824**	−0.0290	−0.0525	−0.0704	0.0345
ROA	0.0049	−0.0538	**−0.8881**	−0.0219	0.0235
OCF	−0.0594	0.1077	**−0.8862**	−0.0764	0.2300
UE_NEG	0.1285	−0.1830	**0.4398**	−0.0102	0.3099
LOSS	−0.0525	0.1356	**0.6937**	−0.0758	0.1627
SPECITM	0.0705	0.1328	0.0450	0.3261	**0.6413**
LIT	−0.0116	**0.6836**	0.0672	−0.1463	−0.1242
VOLUME	0.3905	**0.5923**	−0.1105	−0.0677	0.2194
STDRET	−0.2821	**0.4234**	0.2184	−0.2876	0.2267
LNBSEG	−0.0673	−0.3753	0.0023	**0.6681**	0.2095
MBR	0.1063	0.1569	0.1764	0.0919	**−0.6377**
MKTCONC	0.2017	**−0.4417**	−0.0883	−0.2600	0.0973
Variance Explained	20.08	9.94	18.39	13.71	7.63
Sample 2001 to 2011	23,134	23,134	23,134	23,134	23,134

The above loadings are based on principle component factoring with promax (oblique) rotation. The sample prior to the R-Student restriction is used to generate the factor loadings. Factor loadings greater than 0.40 in absolute value are in bold.
Variable definitions can be found in Appendix A, with further discussion of the factor scores in Appendix C.

return volatility during the year (*STDRET*), and negatively on market concentration (*MKTCONC*). Prior research suggests that class action lawsuits are more likely for companies with greater trading volume and price volatility, providing the necessary stock price decline after unfavorable news to make the case that there was a material misrepresentation that resulted in damages to the investor (Lang and Lundholm 1993).

The third factor, which I label Bad News (*BADNEWS*), loads positively on measures capturing whether the company experienced a net loss during the current year (*LOSS*) and missed earnings expectations (*UE_NEG*). It also loads negatively on measures of return on assets and operating cash flows. Prior literature associates *LOSS* and *UE_NEG* with short-term bad news, which may cause companies to delay or limit the content of their earnings announcement. *ROA* and *OCF* are measures of long-term growth prospects that are associated with increased disclosures (Lang and Lundholm 1993).

The fourth factor, which is labeled Complexity (*COMPLX*), loads positively on measures of size (*LNMVE*), number of shareholders (*SHAREHOLDER*), and complexity (*LNBSEG*), which appears to capture information demand and complexity. The fifth factor (*DFACT*) loads positively on an indicator for special items (*SPECITM*) and negatively on the market-to-book ratio (*MBR*), which has been used as a measure of disclosure complexity, information environment, and proprietary cost.

The measure for proprietary cost (*MKTCONC*) loads negative in the Litigation Risk factor, consistent with the factor loadings in D'Souza et al. (2010), but does not appear to have a definitive separate factor capturing the construct of proprietary cost. D'Souza et al. (2010) point out that proprietary costs may be less important for companies that have already made the decision to issue an earnings announcement, as most proprietary cost models measure the decision to disclose their private information conditional on potential costs associated with competitors and other third-party groups.

American Accounting Association